Fairness Auditing and Bias Mitigation in ML: A Technical Workflow

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ABSTRACT

As AI and ML applications become more common in public organisations, technical processes are needed to audit model behaviour, document limitations, and mitigate measurable disparities before deployment. This study presents a workflow for fairness auditing and bias mitigation in supervised classification, with emphasis on error disparities across groups rather than a complete solution to AI ethics. The workflow is illustrated with two experiments based on data from Chile's Public Criminal Defence Office (DPP), where the task is to predict whether a criminal case has a favourable or unfavourable outcome for the defendant. Each experiment compares a base model with an improved model that uses reweighting, class weighting, threshold optimisation, interpretability analysis, and documentation. Fairness is operationalised through group-level error metrics, especially false negative rate (FNR), false omission rate (FOR), and true positive rate (TPR), using Region as a geographic proxy attribute. The results show that mitigation can reduce selected disparities, but also reveal a clear fairness–performance trade-off. In the drug trafficking experiment, the manuscript-level confusion matrices show a reduction in FNR from 40.54% to 31.63%, with an increase in FPR from 0.13% to 39.25%. Statistical comparisons are computed from the confusion-matrix counts reported in this manuscript and are therefore reported as aggregate two-proportion tests rather than paired McNemar tests. The workflow should therefore be read as a technical audit and mitigation procedure, not as evidence that the resulting models are suitable for operational legal decision-making.

1 Introduction

As Machine Learning (ML) and Artificial Intelligence (AI) tools are more common in our society, new challenges arise when these get applied in the public sector – such as in organisations and/or public policies[12,32]. Since these tools are often considered "black boxes", it is quite difficult to understand exactly how an algorithm arrives at a particular decision. This paper addresses that problem through a technical workflow for auditing and mitigating measurable model disparities, rather than through a general theory of ethical AI.

Ethical issues may arise when considering the inherent biases in the data used to train these algorithms. If the judicial system has historically exhibited bias towards particular demographic groups, an algorithm trained with such data may reinforce or intensify this bias.

The problem is, this can affect people. Going back to the judicial system example, if it has a model that can predict the outcome of a trial, that could influence the final sentence of the defendant, potentially changing their life forever. Of course, not every application of this technology is the same - our social media feed may show us something we do not really want to see, but it would not have a huge impact on our lives.

However, if such technology is implemented in crucial sectors such as healthcare, finance or law enforcement, the ramifications can be considerable and harmful. In these industries, an erroneous forecast or partial verdict could result in an incorrect diagnosis, economic adversity or false imprisonment. It is not exclusively about receiving inaccurate recommendations; rather, it concerns the compromise of fundamental rights, fairness and impartiality.

The use of ML applications in public contexts also raises questions about liability and consent. Who is responsible when an algorithm makes a harmful decision? How can we ensure that these models do not pose risks to our society? How can we be sure that the individuals affected by these decisions were properly informed about the use of such tools?

In fact, one way to address these ethical concerns is through regulation. In June 2023, the European Union approved its first AI regulatory act, which sets out harmonised rules for AI systems. The regulation outlines specific bans on certain AI practices, mandates for high-risk systems, and emphasises transparency in AI-human interactions and media manipulation. It also provides a comprehensive framework for market surveillance, covering both providers and users of AI systems in the EU[50]. Governments are not the only ones interested in this, industry is too. OpenAI wants regulation for what it calls “frontier AI”[38], i.e., models whose capabilities could pose serious risks to public safety.

In addition, AI cannot replicate many human capabilities. In the field of law, a lawyer must combine abstract thinking with skills to solve problems in situations of high uncertainty, in both the legal and factual domains; something a classification model cannot yet achieve[9]. That is, models cannot ~~“reason”~~ “reason” in the same manner as a human. Even the large language models ~~—~~ – such as OpenAI’s ChatGPT or Google’s Gemini ~~—~~ – that have become so popular recently are not capable of doing so, or at least not independently[39,40]. ~~This is due to the fact that behind that black box, there are only probabilities and computations among vectors and matrices. Hence, models should be “ethically trained” in order to avoid these problems~~Classification models should therefore be used, if at all, only within carefully documented decision-support, auditing, or quality-control workflows.

It is crucial to recognise that while the potential benefits of AI and ML are ~~enormous~~considerable, the risks they pose, especially if left unchecked, can have profound implications for individuals and society as a whole. This is especially important in legal settings, where historical labels can encode institutional practices and where a favourable or unfavourable outcome label should not be confused with a complete account of justice, legal merit, or defence quality.

1.1 Background

Several studies have suggested various guidelines and methodologies for AI ethics[43–46,56], but they have not yet converged to a standard in technical and ethical terms[34–37], as many of these studies are task-specific and focused on AI development rather than on ethical considerations[41,42].

However, ethical assessment tools can serve several purposes, including identifying disparities and biases in models. These tools, such as What-If-Tool[22], SHAP[29], AI Fairness 360[30], Fairlearn[31], and Aequitas[14], are effective for analysing model performance, optimising models, and auditing their results. For documentation purposes, there are different tools available such as Model Cards[13] and Datasheets for Datasets[23], among others.

The problem is, the use of ethical tools and techniques for ML algorithms is not widespread when developing such models. This can be explained by the fact that there are no standardised processes to apply these tools[55], and when such processes do exist, they are often seen by AI/ML developers as an extra burden rather than a tool for facilitating accountable employment of ML models[28,54]. Additionally, their recent emergence and a lack of obligation for their use by both public and private organisations means that there is limited awareness of the importance of employing these tools to address ethical issues.

Moreover, in the public sector, the use of these tools becomes even more critical. Given the public sector’s responsibility to ensure fairness, transparency, and accountability, employing ethical tools to evaluate ML models is essential to build trust and prevent biases that could impact society at large.

For instance, a common use case is in models that classify taxpayers, debtors, and financial and tax-related issues. Specifically, Black et al. (2022)[24] conducted a study on income fairness in the tax audit models used by the US’ Internal Revenue Service (IRS). The study examined vertical equity, that is, how much the model represents important individual differences, given their relevance to public tax policies. Using various ethical tools, the study discovered that prioritising return on investment (ROI) from auditing lower-income taxpayers by the IRS compromises vertical fairness in an effort to decrease the expenses associated with such audits.

Healthcare is another crucial field for ML models. Rajkomar et al. (2018)[27] investigated the effects of these algorithms on medical care, particularly how biases and disparities could put more vulnerable groups at risk of incorrect diagnoses or treatments. The research employed distributive justice strategies to guarantee impartial outcomes for patients and ensures that resource allocation is as equitable as possible.

Singh & Joachims (2018)[26] proposed a methodology to ensure fairness in rankings. The study emphasized the prevalence of rankings in various contexts, including books, movies, jobs, and individual comparisons. Therefore, it is essential for classification algorithms to prioritize not only the benefit of the end-user but also the fairness of the ranked entity. The research has presented various probabilistic approaches that act as constraints in diverse situations, like ensuring demographic parity or addressing disparities in data.

Ethical algorithms encompass not only the models’ operations but also their documentation, specifically their transparency. Heger et al. (2022)[28] examined the needs and perspectives of individuals constructing ML models on data documentation, using datasheets for datasets. The study uncovered the absence of standardised methodologies for comprehensive and user-friendly documentation. Furthermore, there remains a disparity in users’ attitudes towards documentation, often seen as an extra burden rather than a tool for facilitating accountable employment of ML models.

Data also have important implications for model training. According to a recent study on the use of ML models to detect

COVID-19 in X-ray images from various databases[33], researchers discovered that data imbalances and biases in patient age and sex can lead to underperformance of the algorithm, which affects the accuracy and generalisability of predictions.

Furthermore, Davidson et al. (2019)[48] determined that numerous Twitter/X datasets exhibit biases towards African American English speakers/writers. This is particularly problematic when applied in technologies for detecting abusive language, as these systems tend to classify these tweets as abusive more frequently than those written in standard English, resulting in discrimination against groups who are often the targets of such language abuse. According to Wiegand et al. (2019)[47], these models' biases are closely related to the way the data was sampled.

1.2 Key Definitions

Several terms have become crucial when considering the ethical and practical implications of models and their decision-making capabilities. These terms shed light on how models process information, how they can potentially introduce inequalities, and how to interpret their results. Some of these fundamental concepts that are going to be used in this paper are as follows,

1. **Transparency:** This refers to the comprehensibility of an algorithm's decision-making process. In essence, algorithms should not operate as "black boxes" where decisions are made without clarity. Ideally, we should have a clear understanding, or at least a basic idea, of how the algorithm works.

2. **Bias:** This refers to any systematic bias in the algorithm's results that favours certain groups over others. For example, an algorithm that consistently predicts that people from a certain demographic group are likely to underperform in a job - regardless of reality - is biased.

3. **Fairness:** ~~This ensures that each group of people is treated fairly by the algorithm. In practical terms, the results of the algorithm should not unfairly discriminate against any particular~~ In this paper, fairness is treated as an empirical property of model errors across groups, not as a complete normative judgement. We focus on whether error rates such as FNR, FOR, and TPR differ substantially by group.

4. **Disparities:** This term addresses ~~the~~ differences in outcomes or ~~effects that models produce across various demographic groups.~~ error metrics across groups. For example, if defendants from one region have a much higher FNR than defendants from another region, the model misses favourable outcomes for that region more often. Disparities can arise even in the absence of explicit bias, highlighting the nuanced nature of fairness. ~~It underscores the importance of evaluating the broader impacts of algorithms across different sections of the population to ensure equitable results.~~

5. **SHAP Value**: Comes from SHapley Additive exPlanations. Is a game-theoretic approach to explaining the outputs of ML models. Inspired by Shapley values, they assign the contribution of each feature to a prediction, ensuring a fair distribution. This provides a better understanding of model interpretability and individual predictions[51].

1.3 Research Context & Paper Structure

This paper was produced as part of the Ethical Algorithms (*Algoritmos Éticos*) project, led by the GobLab of the University Adolfo Ibáñez and developed in collaboration with the Inter-American Development Bank (IDB), which aims to incorporate ethical standards into the development of tools based on ML and AI models. These ethical standards aim to improve the levels of transparency, fairness, privacy, explainability, and accountability of ML models. It also seeks to establish new purchasing standards for these tools in public tenders.

This project focuses primarily on those performed by or for public institutions and/or state agencies. In particular, this paper is part of the work carried out for the Public Defence Office (DPP) of Chile, a public body that provides defence services to anyone who does not have a lawyer to defend them in a criminal trial. The DPP needed a predictive system to help predict the outcome of a trial in order to carry out audit and quality control tasks on its defence lawyers.

The results shown in this paper correspond to the application of a ~~framework~~ technical workflow developed for this purpose, which is used to evaluate inequalities and biases in ML models. This ~~framework aims not only~~ workflow aims to help analyse~~and understand these problems , but also to mitigate them~~ , document, and partially mitigate these problems by using different tools~~to optimise and address them~~, but it does not claim to solve the broader ethical and legal questions raised by predictive systems in criminal justice. The methodology was put into test by doing two experiments, each one with two models: a base model ~~with no ethical considerations,~~ and an optimised one. ~~In the end, both models were improved in both disparitiesand biases, whilst having a minimal impact on performance~~ The results show reductions in selected error disparities, but also show that mitigation can reduce conventional performance metrics or increase other error types. Our best results include a false negative rate reduction from 40.54% to 31.63%, and ~~its disparities between attribute~~ disparities between attributes reduced by an average of 44.37% for one of the experiments.

This paper is structured as follows. Section 2 presents the proposed methodology for this project, as well as the datasets and tools that were used. In section 3, the experiments, models and results are shown; whilst in section 4 those results are discussed. Finally, section 5 contains the conclusion and future work.

2 Methodology

We propose a ~~comprehensive framework~~ technical workflow for evaluating and enhancing ML models in terms of fairness and bias, applicable across different techniques like gradient boosting machines and neural networks. This ~~framework~~ workflow is designed to answer key questions on model performance, ~~improvements, and~~ error distribution, and model functioning in the context of fairness ~~and bias~~auditing.

It employs a two-part experiment format, contrasting a base model with an ~~ethically~~ improved model that addresses ~~fairness and biases. The framework~~ measurable fairness and bias concerns. The workflow emphasises pre- and post-training phases, and encompasses elements of analysis and documentation for ~~a complete ethical model evaluation~~an applied model audit.

The proposed framework follows a five-step process,

1. Perform basic Exploratory Data Analysis (EDA) to identify data characteristics, potential disparities, and class imbalances.

2. Evaluate the base model regarding its performance, biases, and fairness (disparities).

3. Using the information obtained, develop an improved model factoring in ~~fairness and ethical~~ fairness-related considerations, including mitigating potential disparities, balancing sample and class weights, optimising hyperparameters, and selecting the optimal classification threshold.

4. Evaluate the newly created model and compare it to the base version.

5. Apply documentation frameworks and other tools to enhance transparency and understanding of the model's workings and decision-making process.

2.1 Problem description

As it was previously mentioned, the problem comes from Chile's Public Criminal Defense Office (*Defensoría Penal Pública*, DPP). This entity needed a tool based on ML techniques that could help in predicting the outcome of a trial, so that they could conduct a series of audits and quality control tests on its lawyers.

Since this implies the use of sensitive data from people processed by different crimes, it requires some kind of ethical analysis of the models developed for this purpose, ~~specially~~ especially since they are trying to predict an outcome that could potentially ~~change the life of the~~ affect the defendant. In particular, every model developed has one class to predict: if a person has a favourable (1) or unfavourable (0) outcome to its trial. An unfavourable outcome occurs when the defendant is imprisoned and/or its sentence gets increased. A favourable outcome is anything but the ~~previous~~ previously mentioned – i.e.: sentence reduced, paroled or acquitted. This binary labelling is a practical simplification of legal outcomes and has important limitations: it does not determine whether an outcome was legally correct, whether defence quality was adequate, or whether historical judicial patterns were themselves unbiased.

One of the main concerns that can arise is the way in which the model(s) handles the predictions. This is because of the potential biases that may arise due to the nature of the crime - i.e., its prevalence in certain areas of the country, or by whom it is committed. In that regard, it is important to determine whether one or more models (one per crime) should be developed. A single model would facilitate the development process, but because the crimes may be so different, it may have problems of accuracy, which would mean that it would not be very effective for its purpose. On the other hand, with multiple models, this shall not be a problem, as each would be a specialist in one crime. This can also facilitate ethical analysis, as it allows for the model to inherit any biases and differences that might be found within a given crime.

Similarly, another concern that may arise is the way in which these models are used. While they are intended to be used for auditing purposes, it cannot be ruled out that a lawyer might use them to determine whether or not it is in his or her interest to defend an accused person. It is important to have clear documentation that correctly defines the purpose of these tools, how they work and a clear definition of the potential risks associated with their use.

2.2 Databases

We are using two of the datasets that were handed over for this work. These contain the records of multiple individuals accused of drug trafficking or petty theft in between the years 2017 and 2022. The details are as follows,

- There are 16,690 non-null rows for the drug trafficking set; whilst the petty theft has 39,954 non-null rows.

- Each dataset has a mixture of defendant-related and trial-related columns. The second ones are directly related to the outcome of the trial, so those cannot be used for training purposes. Hence, we are using those related to the defendant: region (geographical location; "Region"), crime stage ("Desarrollo"), hearings ("Audiencias Efectivas") and the Barrister ("Defensor"); and exclusively for the drug trafficking set, there is also the foreigner ("Extranjero") column.

- However, there are many critical variables (such as the age, sex, and previous cases of the defendant, among others) that were not available in these sets. The absence of these features obviously affects the performance that a model can have, but it also limits the capabilities of analysing potential biases and disparities. Because the data come from historical DPP records, the labels and observed features may also reflect prior judicial and institutional practices. Any model trained on these data can therefore reproduce, and potentially launder, historical biases present in the case outcomes.

- As a result, a new variable called "age" was created for each dataset. For both sets, two versions of the age variable were generated: one simulated using a normal distribution and the other using a uniform distribution. The idea is to ~~ensure that~~ check whether the inclusion of this synthetic variable ~~does not skew the final~~ materially changes the model results. This variable is not observed age and should not be interpreted as defendant-level demographic truth.

- For the drug trafficking one, we use the information from the Fourteenth National Drug Survey in the General Population of Chile, made by the National Service for the Prevention and Rehabilitation of Drug and Alcohol Consumption[53] in Chile, which states that people between the ages of 18 and 34 years old concentrates the majority of the drug consumption in the country. The normal distribution variable was generated with a mean (μ) of 26 (the average between 18 and 34 years) and a standard deviation (σ) of 8; while the uniform distribution variable was generated with a lower limit of 18, representing the legal age of adulthood, and an upper limit of 65 years.

- For the second dataset, we use the data from the Centre for Crime Studies and Analysis[52] for the crime of theft between the years 2017 and 2022. The normal distribution variable was generated with a mean (μ) of 31 (average age between 18 and 44, which accounts for the majority of those charged with this crime) and a standard deviation (σ) of 8; while the uniform distribution variable was generated with the same limits from the previous experiment (18 to 65 years).

- There are no visible class imbalances in the first set. However, in the second one there are some important differences. In this case, there are 28,819 unfavourable outcomes, versus 11,135 favourable outcomes. This means that there are 2.5 times more unfavourable outcomes than favourable ones.

2.3 Development & ethical tools

In this subsection, we look at the essential development and ethical tools used throughout our research. Those in the first category are as follows,

1. **Python**: A versatile and widely-used programming language, Python offers extensive libraries and frameworks for ML and data analysis, facilitating the design and implementation of our algorithms[15].

2. **Data processing and visualisation**: This includes libraries such as Pandas[16] and Numpy[17] for data manipulation, analysis, and transformation, providing a foundation for our data preprocessing needs. For visualisations, we use Matplotlib[18] for basic graphs and plots.

3. **Scikit Learn**: An open-source ML library for Python, Scikit Learn provides simple tools for data analysis and modelling, making it easier to implement ML algorithms[19].

Fairness Enhancers:

1. **AI Fairness 360**: An extensible open-source toolkit that offers a comprehensive set of fairness metrics for datasets and ML models, as well as algorithms to mitigate biases.

2. **Fairlearn**: A tool that focuses on assessing and improving fairness in ML models, offering mitigation algorithms and fairness metrics.

3. **Aequitas**: An open-source bias audit toolkit, Aequitas facilitates the process of understanding and interpreting disparities in ML models across various groups.

Analytical and documentation tools:

1. **What-If-Tool**: An interactive visual interface that provides in-depth exploration capabilities for ML models, allowing users to understand their predictions more profoundly.

2. **SHAP**: Offers a game-theoretic approach to model explanations, providing clarity on feature contributions to individual predictions, enabling more transparent model interpretability.

3. **Model Cards**: A framework designed to provide comprehensive and standardised reporting on ML model performances, ensuring transparency in their capabilities and limitations across diverse operational environments and user groups.

2.4 Experiments

Two experiments were conducted, one with the drug trafficking set, and the second one with the petty theft set. Each experiment was divided into two parts: the base situation and the improved situation.

As it was previously mentioned, the purpose of these models is to predict if a defendant will a have a favourable or unfavourable outcome to its trial. Each dataset can have one or more outcomes (e.g., ~~‘~~“Outcome 1~~’, ‘~~”, “Outcome 2~~’~~”, etc.), which represent the results of legal proceedings that may change over time through different stages or instances of the judicial process. For example, an individual might initially receive an ~~unfavorable~~ unfavourable outcome, such as a prison sentence, but later, after an appeal, the outcome might change to a more ~~favorable~~ favourable one, such as house arrest. To simplify the analysis, the models developed in this study will only consider the most recent outcome for each case. For the drug trafficking set, it will be the second outcome; for the petty theft dataset, it will be the first (and only) one.

1. The base situation consists of a basic model that can predict the previous mentioned classes. Once the model is created, the results are analysed using SHAP values[29], the What If Tool[22] and Aequitas[14], so that the biases and disparities get identified.

2. In the improved situation, as its name suggests, the model gets refined upon the insights gathered in the first part. This also considers the selection of a protected feature. In this case, for both experiments, it is going to be the “Region” feature.

   (a) For the pre-training phase, we use AI Fairness 360[30] and its Reweighting tool to assign new weights to the sample data, as well as Scikit-Learn’s[19] *compute_class_weight* function for optimising the class weights. Some basic hyperparameter tuning was also applied for some gains in model performance.

   (b) For the post-training phase, we use Fairlearn’s *ThresholdOptimizer* function for optimising the prediction threshold of our model. The constraint used for this is the false negative rate parity, since we are interested in minimising the amount of people that has a non-favourable ending to its trial. After that, the model gets analysed using the same tools mentioned before. Additionally, it can be documented using Google’s Model Cards framework for model documentation.

For the protected variable, we want to select those regions that are under-represented in the dataset. There are a total of 16 regions in Chile, and since we want to ensure a fairly equal representation of these places, in both experiments the regions will be sorted by number of records (highest to lowest), and half of them will be selected (8) as ~~favored~~ favoured and/or ~~unfavored.~~ unfavoured. Region is used here as a geographic proxy attribute because more conventional protected variables were unavailable. This choice is useful for auditing territorial disparities, but it is limited: Region is not a substitute for race, ethnicity, socioeconomic status, gender, real age, or other individual-level protected attributes.

It is worth mentioning that the selection of models for both experiments was completely arbitrary. Two different architectures were chosen specifically to illustrate that the proposed framework can be applied across various machine learning configurations.

Figure 1 shows a detailed flowchart for the proposed framework that is applied for these experiments.

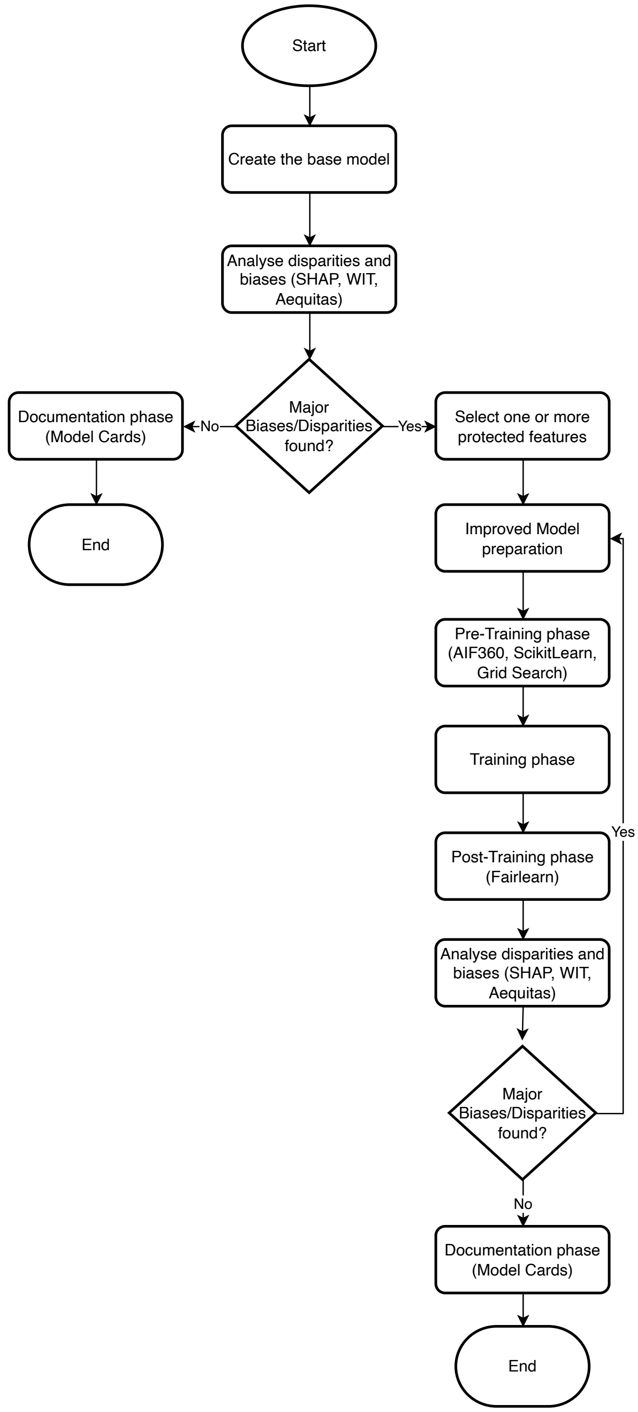

Figure 1. Proposed framework flowchart.

3 Results & Analysis

The following section will only present both experiments using the “age” variable simulated with the normal distribution, as there were no significant differences in the results compared to the uniform distribution. For completeness, the results using the uniform distribution for the “age” variable are included in Appendix A for the drug trafficking experiment and in Appendix B for the petty theft experiment.

To avoid changing the reported experimental results, the statistical tests below use the confusion-matrix counts already presented in this manuscript. These tests should be interpreted as aggregate comparisons, not as paired prediction-level tests. In particular, McNemar tests and bootstrap confidence intervals over paired predictions are not used because those methods require the exact paired predictions corresponding to the reported manuscript run.

The structure of this part is detailed as follows: First, we provide a detailed description of the experimental setup, including the architecture of the models employed, the hyperparameters used, and the modifications implemented in the improved version of the model. This is followed by a comparison of the performance metrics of both models, focusing on accuracy, precision, recall, and F1-score, to quantify their effectiveness.

Next, we analyse the interpretability of the models by examining the SHAP values, which highlight the most influential features in the decision-making process of the models. Then, we evaluate the fairness of the models using bias metrics and the ~~confussion~~ confusion matrix to assess whether any group disparities exist in their predictions. These bias metrics are the following,

- FPR (False Positive Rate): The proportion of actual negative cases that are incorrectly classified as positive.
- FNR (False Negative Rate): The proportion of actual positive cases that are incorrectly classified as negative.
- FDR (False Discovery Rate): The proportion of positive predictions that are incorrect.
- FOR (False Omission Rate): The proportion of negative predictions that are incorrect.

Finally, we analyse the disparities between groups using the Aequitas framework to provide a deeper understanding of fairness and equity in the model outputs. In each experiment, three assessments were made using the different reference groups that Aequitas admits. These are the following,

- The major group for each feature, which is defined as the group with the largest representation in the dataset for a given attribute. For example, if ~~analyzing~~ analysing the attribute Region and “Valparaíso~~"~~” is the most prevalent group, it will serve as the reference group for disparity calculations.
- The group with minimum value for each specified metric or each attribute and metric combination, the group exhibiting the lowest metric value becomes the baseline for disparity calculations.
- A predefined group. In both experiments, it was set to a defendant from the Metropolitan Region (central zone), aged between 18 and 28 years old, Chilean (not foreigner), whose felony was consummated.

These assessments were made considering the True Positive Rate, the False Omission Rate and the False Negative Rate. The TPR is of interest to us to see how positive predictions are distributed within the groups of each feature; FOR is useful to see where the model is failing to make correct predictions; while the FNR aids in understanding the proportion of false negatives over the total actual positives. Finally, our disparity tolerance (τ) is set to 1.5 times larger or smaller than the size of each reference group, which is slightly higher than the threshold recommended by Aequitas (1.25). This choice was made due to the limited amount of data available, and, as there are imbalances in the groups within the dataset, a smaller value could inherently skew the metrics and complicate the analysis.

3.1 Drug trafficking experiment

For this experiment, a multi-layer perceptron was used. The details of the architecture and training for the base situation are as follows,

- **Architecture**: Multi-Layer Perceptron (MLP), developed with Tensorflow, and trained to minimise the binary crossentropy.
- **Type**: Classification. In this case, to predict if a person has a favourable outcome (1) or not (0).
- **Datasets**: 70% split for the training set, 30% for the testing set, and 1000 rows were removed from the training dataset to create a validation dataset. In total, there are 10683 rows for training, and 5007 for testing.

Experiment	Metric	Improved - Base	95% CI	p-value
Drug trafficking	Accuracy	-13.40 pp	[-15.15, -11.65] pp	< 0.001
Drug trafficking	Recall / TPR	+8.91 pp	[6.35, 11.47] pp	< 0.001
Drug trafficking	FNR	-8.91 pp	[-11.47, -6.35] pp	< 0.001
Drug trafficking	FPR	+39.12 pp	[37.13, 41.11] pp	< 0.001
Drug trafficking	F1 score	-6.97 pp	–	–
Petty theft	Accuracy	-2.80 pp	[-3.90, -1.71] pp	< 0.001
Petty theft	Recall / TPR	+9.04 pp	[6.80, 11.27] pp	< 0.001
Petty theft	FNR	-9.04 pp	[-11.27, -6.80] pp	< 0.001
Petty theft	FPR	+7.26 pp	[6.43, 8.08] pp	< 0.001
Petty theft	F1 score	+4.29 pp	–	–

Table 1. Aggregate statistical comparison between base and improved models using the manuscript confusion matrices. Percentage-point differences are improved minus base. Accuracy, recall, FNR, and FPR use two-proportion z-tests; F1 is reported as a descriptive aggregate difference because it has no simple two-proportion test under these counts.

- **Hyperparameters**: Found through manual search.
  - 70 training epochs.
  - Learning rate: 0.01
  - Batch size: 1024.
  - Layers: 4 - One input layer, two hidden layers and one output layer.
  - Neuron configuration: 6, 13, 5, 1.
  - Activation functions: Sigmoid.

For the second part of this experiment, the improved situation, there is a similar configuration with a few changes, those being the amount of neurons for the hidden layers: 14 for the first one, and 12 for the second one; as well as both hidden layers now using a ReLU activation function. These changes, made for testing purposes, were part of an iterative model improvement process, and the updated hyperparameters were also found through manual search.

This new model also uses the improved sample weights that were calculated by AIF360's reweighting function. The class weight were also balanced using Scikit-Learn's *compute_class_weight* method, even though it was not really necessary to do so.

3.1.1 Model performance

Model/ Dataset	Accuracy	Precision	Recall	F1 Score
Base Train	80%	100%	62%	77%
Base Test	78%	100%	59%	75%
Improved Train	67%	67%	72%	70%
Improved Test	65%	67%	68%	68%

Table 2. Model performance.

There are clear differences between the base and improved configurations. As it can be seen in Table 2, the base model excels in accuracy and precision, but the recall is quite low, indicating a tendency towards negative predictions. The improved version, on the other hand, opts for a compromise, favouring a balanced distribution of false negatives and positives, although at the expense of some accuracy and precision. The aggregate comparison in Table 1 supports this interpretation: recall increases by 8.91 percentage points and FNR decreases by the same amount, while accuracy decreases by 13.40 percentage points and FPR increases substantially.

Feature	Model	Avg SHAP	Avg Abs SHAP
Effective Hearings	Base	-0.000126	0.001095
	Improved	0.000162	0.001520
Age (Normal)	Base	0.000398	0.003312
	Improved	0.000034	0.002067
Region	Base	-0.000790	0.008457
	Improved	-0.000052	0.006592
Barrister	Base	-0.001273	0.014591
	Improved	0.001620	0.025126
Crime Stage	Base	0.000203	0.000203
	Improved	0.000164	0.000164
Foreigner	Base	-0.025131	0.311636
	Improved	-0.027641	0.348416

Table 3. Average and average absolute SHAP values for the base and improved models.

3.1.2 SHAP Values

SHAP values were computed using the training dataset for both the base and improved models to evaluate feature importance and their contribution to model predictions. Table 3 provides a summary of SHAP metrics for each feature, including the average SHAP value (indicating whether the feature tends to contribute positively or negatively to predictions) and the average absolute SHAP value (highlighting the overall magnitude of the feature's impact on the model).

The base model SHAP values, according to Table 3, shows that the variables "Foreigner" and "Barrister" have the most significant influence on the model's outputs, trailed by the region, the age, the effective hearings, and finally, the crime stage. When looking at the average SHAP values, it can be seen that the first three most influential ~~varibles~~ variables (foreigner, barrister and region) tend to have a negative impact on predictions – with an average of -0.025131, -0.001273 and -0.000790, respectively, as well as "~~Effecive~~ Effective Hearings", with ~~and~~ an average value of -0.000126. Only "Age" and "Crime Stage" have on average positive values, although they have a negligible impact on the predictions.

In the improved configuration, both "Foreigner" and "Barrister" have a more pronounced influence in the model, as their absolute SHAP values went up on average from 0.311636 and 0.014591, to 0.348416 and 0.025126, respectively – which is a 11.8% increase for the variable "~~Foreinger~~Foreigner", and a 72.20% for "Barrister". The first variable still has an overall negative impact on the predictions (as it has an average value of -0.027641, a slight increase from the base model), while the second one now has a positive influence, with its average value ~~is now~~ now at 0.001620, up from -0.001273 from the baseline configuration.

For the variable "Region", the average absolute SHAP value went down from 0.008457 to 0.006592 (a 22.06% decrease), which also means that its average SHAP ~~also~~ went down, from -0.000790 to -0.000052~~, which~~. This makes this variable less influential in the model~~'~~'s outcomes compared to its previous version~~. This suggests that predictions are less likely to be biased due to factors related to the defendant's region, thereby enhancing the fairness of the model. This suggests that predictions are less likely to be biased due to factors related to the defendant's region, thereby enhancing the fairness of the model~~, although it does not prove that regional bias has been removed.

Figures 2 and 3 should be interpreted as follows,

- The X-axis represents the real SHAP value and its effect on predictions.
- Negative SHAP values, on the left side of zero, indicate that the variable contributes to a negative prediction by the model.
- Positive SHAP values, on the right side of zero, indicate that the variable contributes to positive predictions.
- The feature value is indicated by the colour, where blue denotes lower values and red higher ones.

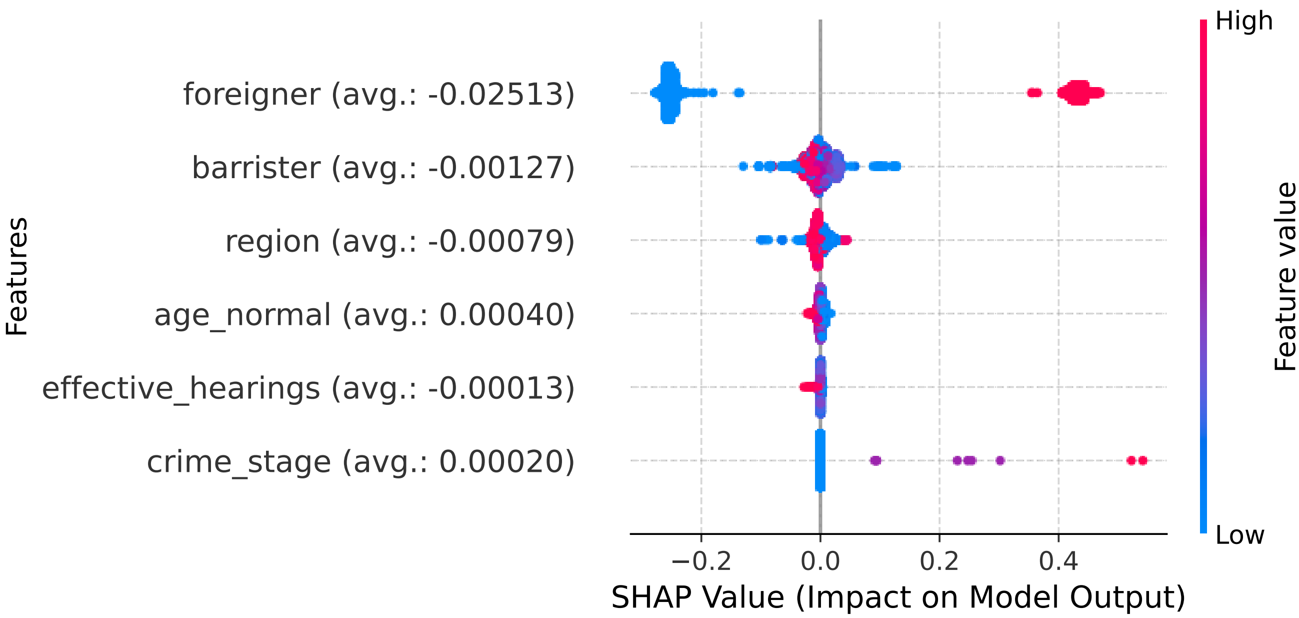

Figure 2. SHAP values distribution by feature (base model).

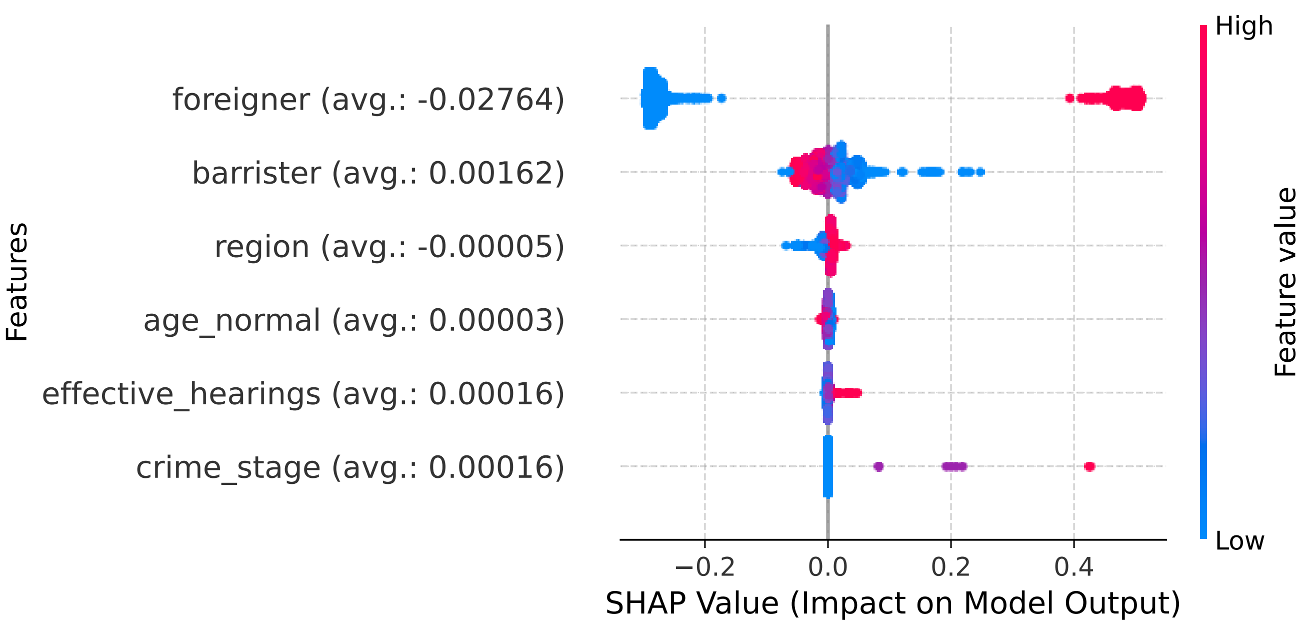

Figure 3. SHAP values distribution by feature (improved model).

These two figures ~~complements~~ complement what was shown in Table 3, which reveals the dual nature of the “Foreigner” variable, which can either positively or negatively affect the model~~’~~’s output. In the base model, lower value regions ~~leans~~ lean towards negative predictions, a likely reflection of overrepresentation of certain regions, especially from the northern areas. However, as it was discussed previously, the improved model mitigates this imbalance, leading to a more even influence of this variable on predictions. Similarly, the ~~“Barrister”~~ “Barrister” variable shifts in its impact, transitioning from a negative effect in the base model to a neutral yet slightly positive influence after optimisation.

3.1.3 Bias and Confusion Matrix

For this part of the experiment, a new column was added for analysis purposes. This is “~~Zona~~Zone”, or the zone of the country, as Chile can be divided ~~intro~~ into three different zones according to its region: north, central and south. This can be useful for analysing the impact of the model in a broader way than “Region”.

The overall metrics for the False Positive Rate (FPR), False Negative Rate (FNR), False Discovery Rate (FDR) and False Omission Rate (FOR) between attributes are shown in Table 4. The confusion matrix for both models can be found on Table 5.

Model	FPR	FNR	FDR	FOR
Base	0.13%	40.54%	0.19%	31.88%
Improved	39.25%	31.63%	33.25%	37.51%

Table 4. Bias metrics between attributes.

Model	True Positives	True Negatives	False Positives	False Negatives
Base	1594	2323	3	1087
Improved	1833	1413	913	848

Table 5. Confusion Matrix.

Insights from Table 4 highlight the base model’s proficiency in positive predictions but its struggle with negative ones. Specifically, its high FNR stands out. The improved version addresses this but experiences an uptick in the FPR ~~—~~ – from 0.13% to 39.25% ~~—~~ – in contrast to a decrease in FNR, from 40.54% to 31.63%. Table 5 complements this, as the number of true positives went up from 1594 to 1833, while false negatives were reduced from 1087 records to 848 – an approximate 22% reduction; by contrast, the amount of false positives is roughly 304 times bigger than in the base model, as ~~the~~ it rose from 3 to 913 records.

3.1.4 Disparities (Aequitas)

The following ~~Figures~~ figures use a combination of ~~colors~~ colours and bar lengths to visually convey the results of disparity assessments for each group across the attributes. Green dots/bars indicate that the group~~'~~’s metric is within the acceptable disparity threshold when compared to the reference group, signifying a ~~“pass”~~ “pass” in the parity test. Red dots/bars denote that the group~~'~~’s metric exceeds the disparity threshold, indicating a ~~“fail”~~ “fail” in the parity test. ~~Gray~~ Grey dots/bars represent the reference group itself, serving as the baseline for comparison. Each dot corresponds to an individual group for a specific attribute, while the length of each bar reflects the proportion of groups that either pass, fail, or are the reference group.

Table 6, as well as Figures 4, 5 and 6 are the assessment results for the base model; while Table 7 is the crosstab for the variable “Region”, broken down by attribute value. Columns “TPR”, “FNR”, “FOR” and “FDR” are the previously mentioned bias metrics.

Assessment	FNR Disparities	FOR Disparities	TPR Disparities
Reference Group	0.81	1.04	21.005
Major Group	7.53	1.61	19.96
Min. Metrics Group	9.15	2.04	28.12

Table 6. Average disparities for the base model.

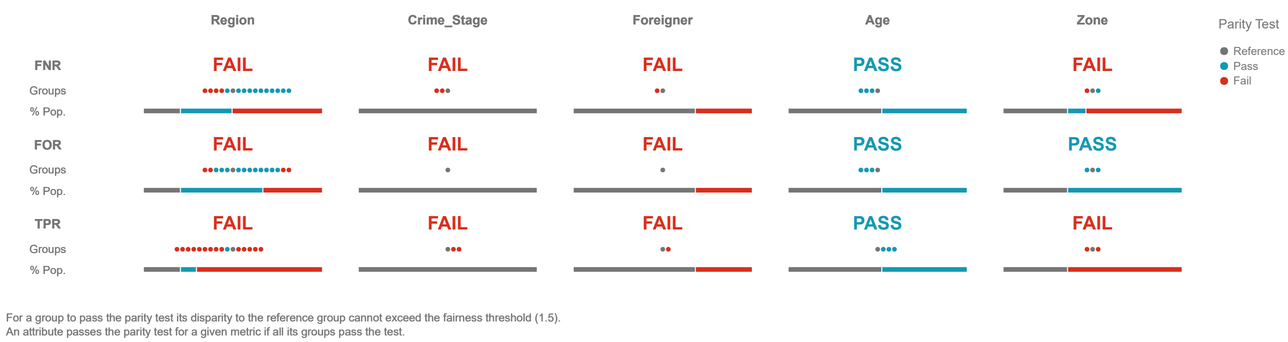

Figure 4. Model assessment for the reference group.

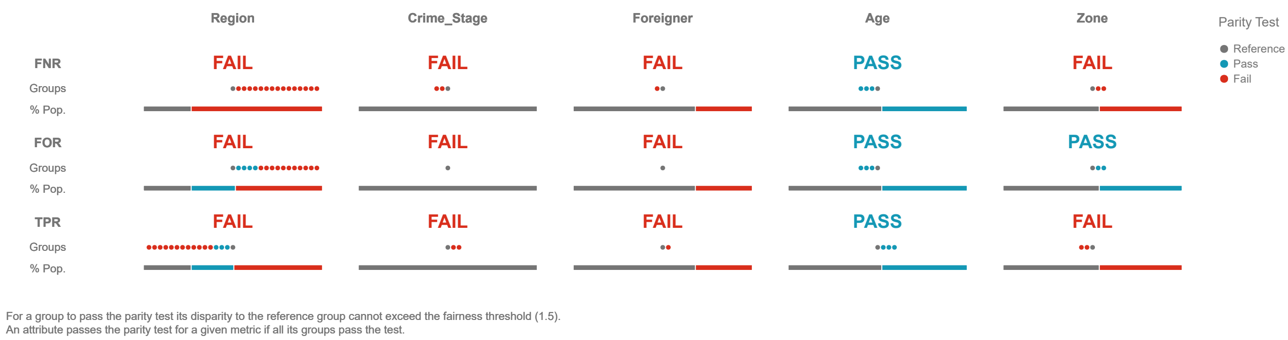

Figure 5. Model assessment for the major group.

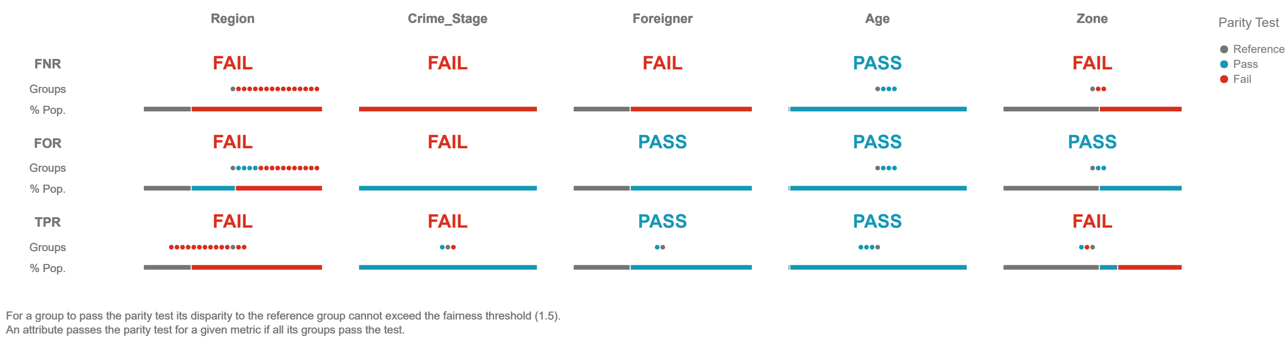

Figure 6. Model assessment for the min metrics group.

Attribute Value	TPR	FNR	FOR	FDR
Antofagasta	0.8213	0.1787	0.2167	0.0000
Arica y Parinacota	0.7276	0.2724	0.2562	0.0000
Atacama	0.6782	0.3218	0.2456	0.0000
Aysén	0.0000	1.0000	0.3889	-
Bio Bío	0.0139	0.9861	0.4057	0.0000
Coquimbo	0.2547	0.7453	0.5852	0.0000
La Araucanía	0.0556	0.9444	0.3238	0.0000
Libertador Bernardo O'Higgins	0.0566	0.9434	0.3759	0.2500
Los Lagos	0.0227	0.9773	0.3554	0.0000
Los Ríos	0.0000	1.0000	0.1818	-
Magallanes y Antártica Chilena	0.0909	0.9091	0.3448	0.0000
Maule	0.0156	0.9844	0.5000	0.0000
Metropolitana	0.1525	0.8475	0.3067	0.0357
Tarapacá	0.9377	0.0623	0.1720	0.0000
Valparaíso	0.1117	0.8883	0.3911	0.0000
Ñuble	0.0833	0.9167	0.4000	0.0000

Table 7. Average bias metrics between attributes for the base model.

In the base model, as evidenced by Table 6 and corresponding Figures 4 to 6, notable disparities emerge. Its performance metrics indicate consistent failure, with pronounced disparities, especially in the latter metrics. The "Foreigner" variable is a lone exception, achieving FOR parity. Focusing on the critical "Region" variable (Table 7), only a few regions meet the benchmark, but the average FNR per attribute is 74.86%, which means that the model tends to incorrectly predict true positives as negative depending on the region of the defendant in 74% of the cases. The average TPR is 25.13%, which means that the model also has problems in correctly identifying true positives. Finally, the FOR is 34%, meaning that of all the negative predictions made by the model, 34% were actually positive and were erroneously missed. When extending the analysis to the "Zone" variable, disparities in FNR and TPR become even more apparent, underscoring significant regional discrepancies.

Table 8, along with Figures 7, 8 and 9 are the assessment results for the improved model. Table 9 is the crosstab for the variable "Region".

Assessment	FNR Disparities	FOR Disparities	TPR Disparities
Reference Group	1.05	1.21	0.99
Major Group	1.06	0.87	0.99
Min. Metrics Group	2.12	2.20	1.78

Table 8. Average disparities for the improved model.

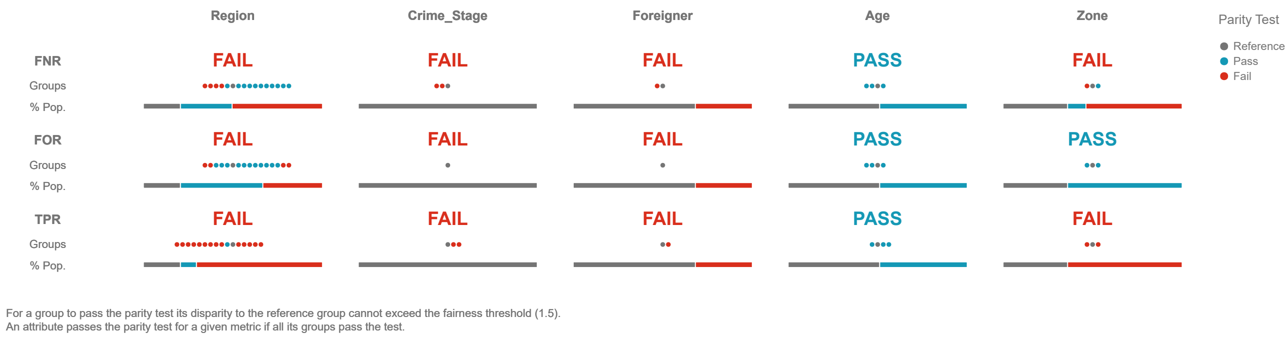

Figure 7. Model assessment for the reference group.

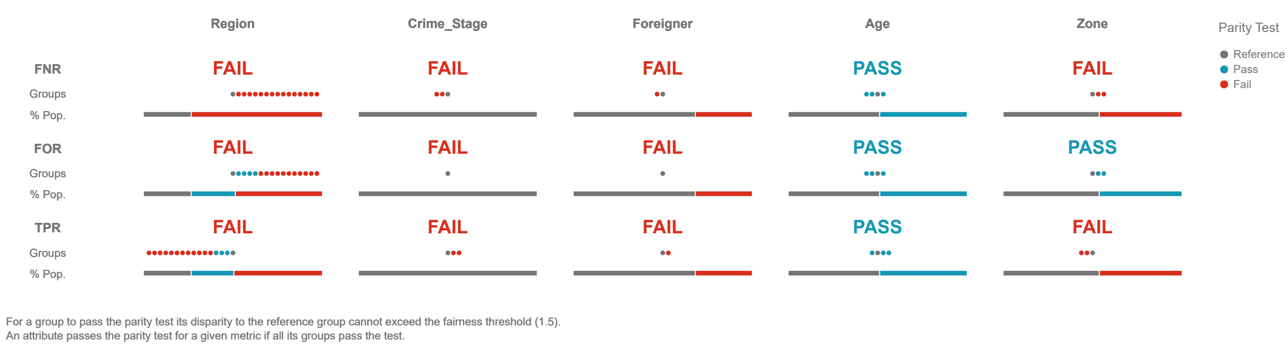

Figure 8. Model assessment for the major group.

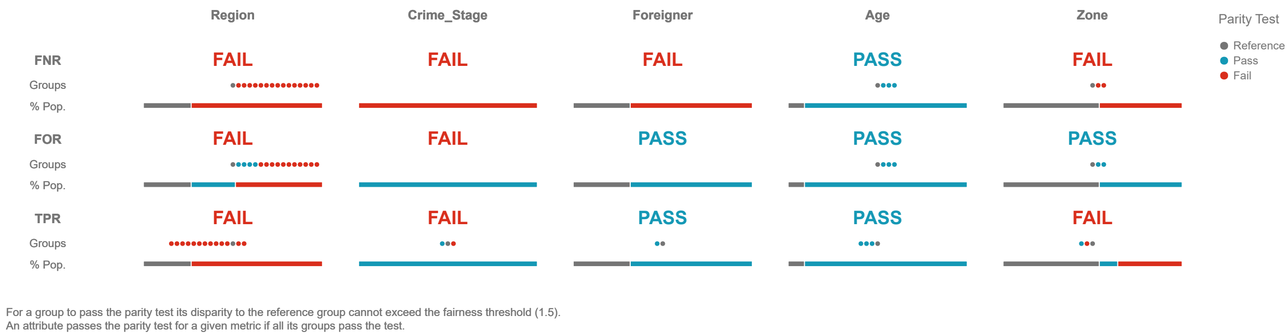

Figure 9. Model assessment for the min metrics group.

Attribute Value	TPR	FNR	FOR	FDR
Antofagasta	0.7113	0.2887	0.3088	0.0000
Arica y Parinacota	0.7276	0.2724	0.2562	0.0000
Atacama	0.7126	0.2874	0.2427	0.1143
Aysén	0.3571	0.6429	0.4091	0.6429
Bio Bío	0.7083	0.2917	0.4286	0.5984
Coquimbo	0.6226	0.3774	0.6154	0.3196
La Araucanía	0.4167	0.5833	0.4286	0.7414
Libertador Bernardo O'Higgins	0.7925	0.2075	0.2895	0.5758
Los Lagos	0.5682	0.4318	0.3800	0.6528
Los Ríos	0.5000	0.5000	0.1818	0.8182
Magallanes y Antártica Chilena	0.2727	0.7273	0.3333	0.5000
Maule	0.5938	0.4062	0.5098	0.5000
Metropolitana	0.6949	0.3051	0.2935	0.6306
Tarapacá	0.6955	0.3045	0.5040	0.0000
Valparaíso	0.6383	0.3617	0.3598	0.5367
Ñuble	0.6250	0.3750	0.5000	0.6150

Table 9. Average bias metrics between attributes for the improved model.

The improved model shows important improvements, as seen in Table 8 and its associated figures (7, 8 and 9). It exhibits a marked reduction in FNR disparities by an average of 44.37%, FOR disparities by 8.50%, and TPR by 94.67%. Although the model has not attained complete fairness or statistical parity, it represents a significant improvement. The "Region" variable, while still problematic, displays improvement as more regions clear the assessment, as its average FNR was reduced to only 39.76%, while the true positive rate was improved from 25.13% to 60.23%. The average FOR was slightly impacted, as its average value went up to 37.75%. Considering the zones, all now align with the set fairness criteria. In terms of parity, the "Foreigner" variable achieves statistical parity, and "Zone" accomplishes FOR and TPR parity.

3.2 Petty Theft experiment

The second experiment has a different model architecture, in this case using a gradient-boosting framework. The details for the base situation are as follows,

- **Architecture**: Light Gradient-Boosting Machine (LGBM), using the *lightgbm* library.
- **Type**: Classification. In this case, to predict if a person has a favourable ending (1) or not (0).
- **Datasets**: 70% split for the training set, 30% for the testing set. In total, there are 28967 rows for training, and 11987 for testing.
- **Hyperparameters**: Found through manual search.
  - Boosting type: gbdt
  - 100 estimators.
  - 63 leaves per estimator (tree).
  - Unlimited maximum depth.
  - Learning rate: 0.01

For the improved model, the amount of leaves per tree goes up to 120 after doing a basic grid search optimisation. It also considers the new sample weights calculated by AIF360, as well as the new class weights from the *compute_class_weight* function.

3.2.1 Model performance

Model/ Dataset	Accuracy	Precision	Recall	F1 Score
Base Train	78%	77%	33%	46%
Base Test	76%	67%	27%	38%
Improved Train	77%	62%	43%	51%
Improved Test	74%	53%	35%	42%

Table 10. Model performance.

The results of this experiment, as described in Table 10, show a suboptimal performance for both models, due to the lack of key variables in the dataset that are essential for effective training. However, the improved model shows a clear advantage ~~, outperforming the base model with superior~~ in recall and F1 values. ~~While~~ This advantage comes with lower accuracy and precision~~show marginal fluctuations, the overall performance of the improved model makes it superior.~~ , so the improved model should not be described as globally superior. The aggregate comparison in Table 1 shows the same trade-off: recall increases by 9.04 percentage points and FNR decreases by 9.04 percentage points, while accuracy decreases by 2.80 percentage points and FPR increases by 7.26 percentage points.

3.2.2 SHAP Values

SHAP values were calculated using the data from the training set for both models. The averages are shown in Table 11.

This table reveals that the base model, on average, predominantly relies on the number of hearings to drive its decisions. The regional aspects and the crime stage come in ~~seconde~~ second place, whereas factors like age and barrister ~~contributes the~~ contribute less. Table 11 also shows that both "Effective Hearings" and "Barrister" have a negative influence towards predictions, as ~~its average SHAP value is~~ their average SHAP values are negative. The improved model displays subtle alterations, with the "Crime stage" variable slightly edging over "Region", as its impact is now 1.33 times bigger than in the base model, while the second variable actually reduced its effect by 2.23%. Additionally, this last variable now has a more neutral effect, as its SHAP value was reduced by 49.05% (0.023921 to 0.012183). The amount of effective hearings is still the dominant variable, having a negligible change compared to the base model (a reduction of 0.77%), and is now 1.09 times more influential towards negative predictions. "Age", "Barrister" and "Crime Stage" have increased their positive influence in the improved model, although the last one still has an overall negative impact on the predictions.

Figures 10 and 11 shows the distribution of these values found in the previous table.

Feature	Model	Avg SHAP	Avg Abs SHAP
Effective Hearings	Base	-0.027219	0.377892
	Improved	-0.029877	0.374996
Age Normal	Base	0.000517	0.002934
	Improved	0.001151	0.012211
Region	Base	0.023921	0.186375
	Improved	0.012183	0.182205
Barrister	Base	-0.018796	0.060601
	Improved	-0.013419	0.086808
Crime Stage	Base	0.010832	0.167486
	Improved	0.014400	0.182265

Table 11. Average and average absolute SHAP values for the base and improved models.

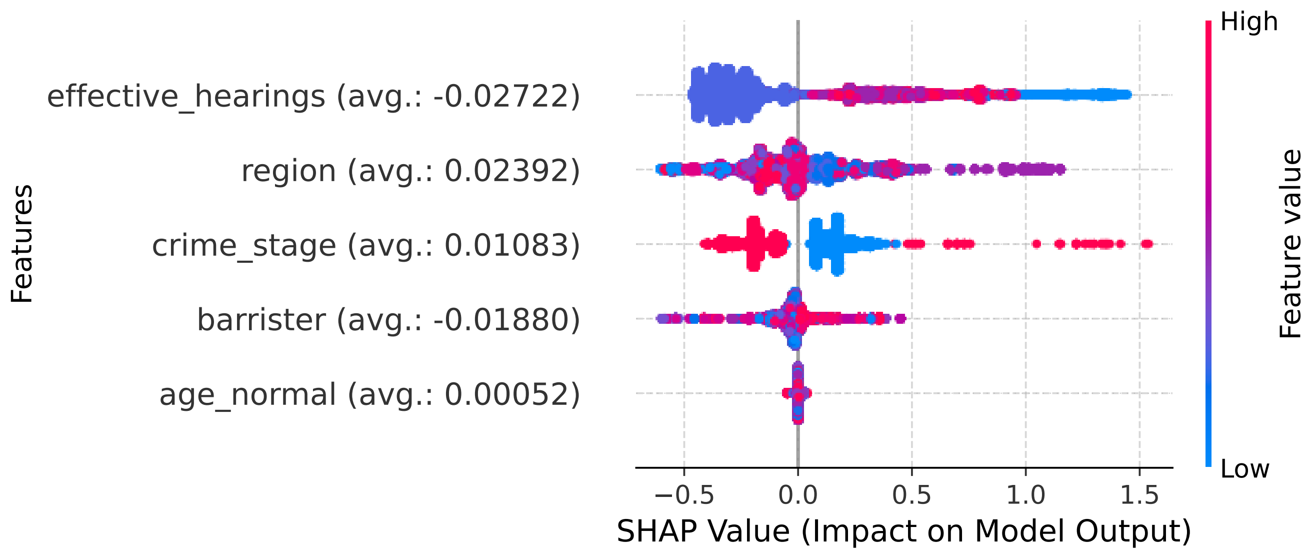

Figure 10. SHAP values distribution by feature (base model).

These figures basically confirm what was discussed previously, although it can be seen that "Hearings" had plenty of cases with big positive effect on the predictions, and it also shows that an increased number of hearings tends to enhance predictive results. As for the variable "Region", there is no visible tendency or skew towards a kind of prediction, independently of the region.

3.2.3 Bias and Confusion Matrix

The same variable – Zone – was added to this experiment. The overall metrics for the False Positive Rate (FPR), False Negative Rate (FNR), False Discovery Rate (FDR) and False Omission Rate (FOR) are shown in Table 12. The confusion matrix for both models can be found on Table 13.

Model	FPR	FNR	FDR	FOR
Base	4.96%	73.20%	32.98%	22.46%
Improved	12.21%	64.16%	47.54%	21.56%

Table 12. Bias metrics between attributes.

Table 12 shows similarities between the situations observed in this and the prior experiment. The base model has problems with false negatives – as its FNR is at 73.20% –, predicting many outcomes as negatives that should have been positives. This manifests in the model's ~~high~~ low recall. The improved model, while attempting to rectify these issues, still struggles with accurate predictions, likely due to the dataset's absence of crucial features needed for training. In this case, the FNR went down to 64.16%, as Table 13 indicates that the number of false negatives were reduced from 2398 to 2102 records (a 12.34%

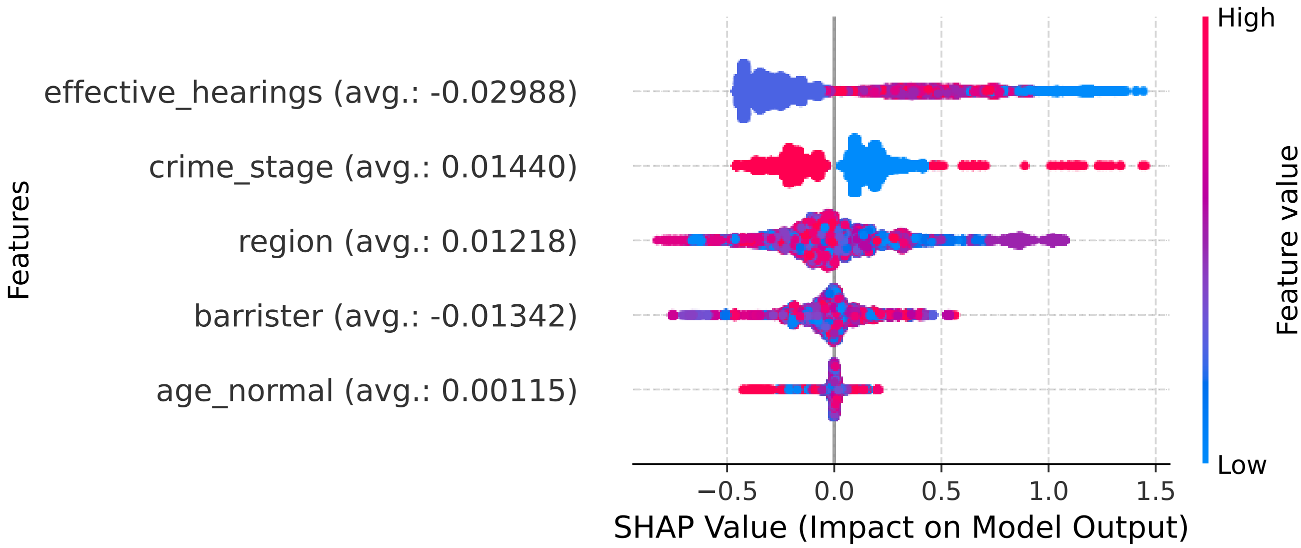

Figure 11. SHAP values distribution by feature (improved model).

Model	True Positives	True Negatives	False Positives	False Negatives
Base	878	8279	432	2398
Improved	1174	7647	1064	2102

Table 13. Confusion Matrix.

reduction). The false positive rate, on the other hand, went up from 4.96% in the base model, to 12.21% in the second one, as the amount of false positives rose from 432 to 1064 predictions (i.e., more than doubled).

3.2.4 Disparities (Aequitas)

The assessments made for this part of the experiment were the same – one with the major group for each feature, the group with the lower measures and a reference group. The metrics used for this task were also the same: FNR, FOR and TPR disparities; and the results for the base model can be found in Table 14 and in Figures 12, 13 and 14. Table 15 is the crosstab for the variable "Region".

Assessment	FNR Disparities	FOR Disparities	TPR Disparities
Reference Group	1.273	1.165	0.644
Major Group	1.273	1.165	0.644
Min. Metrics Group	2.170	2.899	6.272

Table 14. Average disparities for the base model.

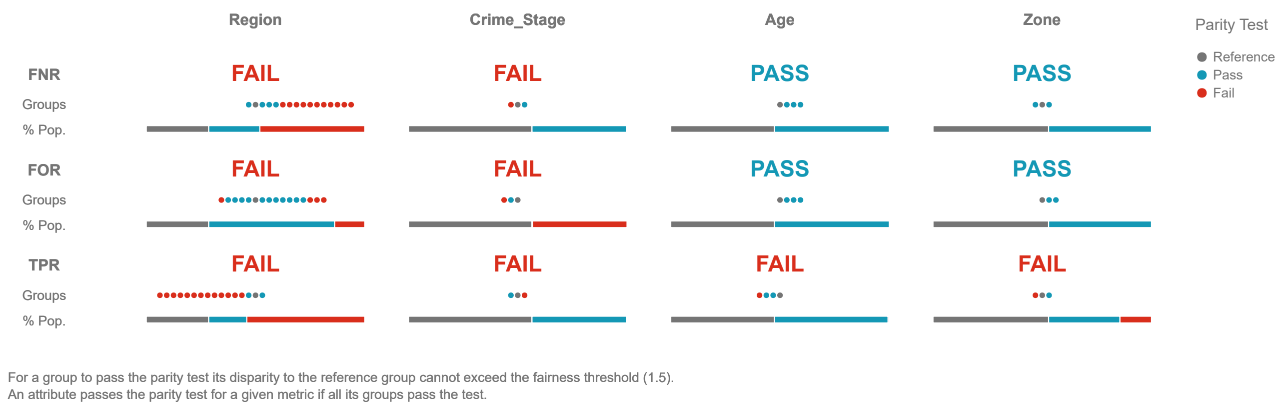

Figure 12. Model assessment for the reference group.

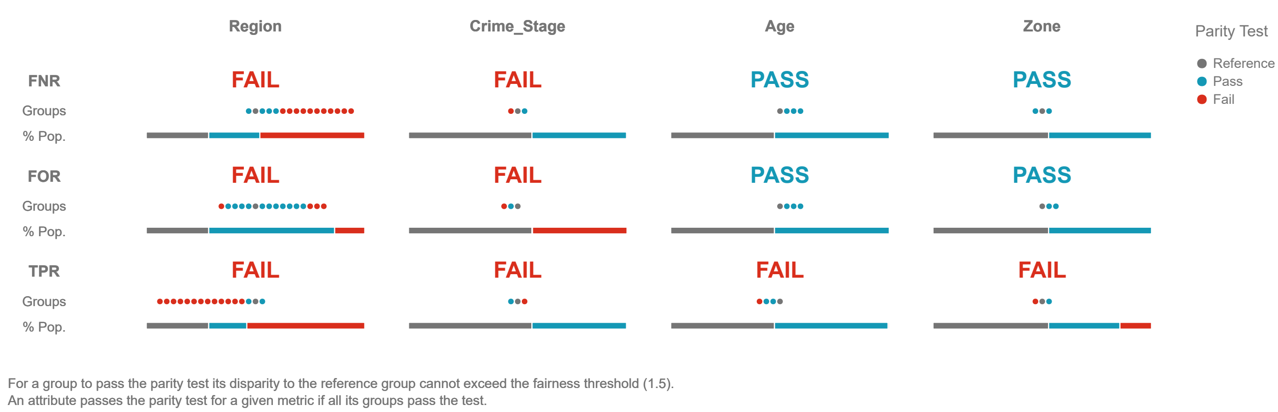

Figure 13. Model assessment for the major group.

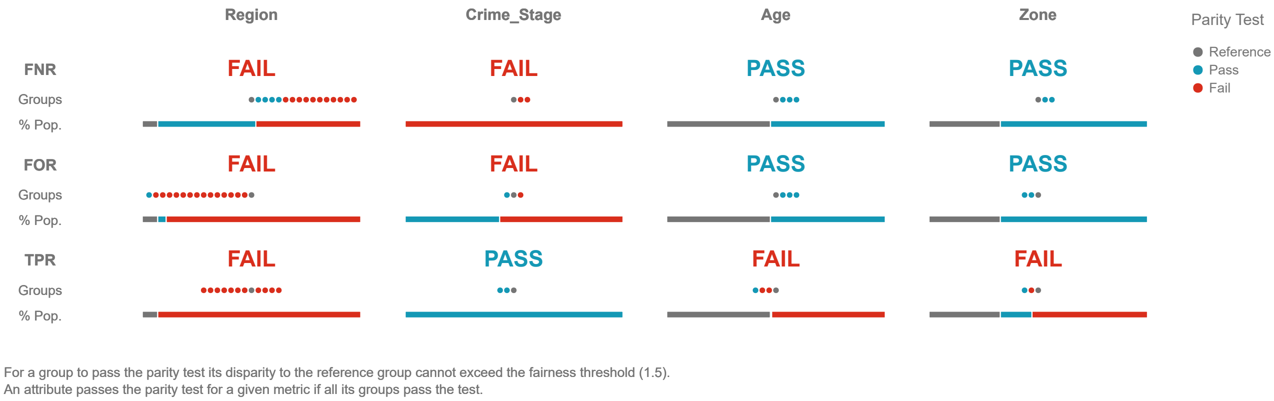

Figure 14. Model assessment for the min metrics group.

Attribute Value	TPR	FNR	FOR	FDR
Antofagasta	0.1235	0.8765	0.4201	0.3333
Arica y Parinacota	0.0806	0.9194	0.3540	0.2857
Atacama	0.2432	0.7568	0.0625	0.4706
Aysén	0.0000	1.0000	0.2700	1.0000
Bio Bío	0.3642	0.6358	0.1750	0.2903
Coquimbo	0.0890	0.9110	0.2180	0.5854
La Araucanía	0.1256	0.8744	0.2815	0.1905
Libertador Bernardo O'Higgins	0.0841	0.9159	0.1943	0.2800
Los Lagos	0.4921	0.5079	0.4945	0.4312
Los Ríos	0.2642	0.7358	0.1674	0.2632
Magallanes y Antártica Chilena	0.0294	0.9706	0.2538	0.8333
Maule	0.0735	0.9265	0.2190	0.1176
Metropolitana	0.4225	0.5775	0.1952	0.2644
Tarapacá	0.0000	1.0000	0.1728	-
Valparaíso	0.0000	1.0000	0.2411	1.0000
Ñuble	0.0000	1.0000	0.2231	-

Table 15. Bias metrics between attribute values for the base model.

Table 14 and related Figures 12 to 14 suggest that the metrics of the base model are slightly better than in the previous experiment. Despite this, it still fails on most tests. However, "Zone" and "Age" achieve FNR parity, and the former also achieves TPR parity. The protected variable "Region" underperforms in many areas. When looking at Table 15, it can be seen that the FNR between attribute values is also high, as its average is at 85%, meaning that more than 8 out of 10 predictions that were classified as negatives were actually positive, denoting big accuracy problems in between regions. TPR is also low, as it is on average at 14.05%, meaning that the model also has problems when classifying positives. The false omission rate, however, stayed relatively low with an average of 24.64%.

Table 16 and Figures 15, 16 and 17 are the assessment results for the improved model, while Table 17 is the crosstab for the variable "Region".

Assessment	FNR Disparities	FOR Disparities	TPR Disparities
Reference Group	1.001	1.075	0.999
Major Group	1.004	1.078	0.993
Min. Metrics Group	1.362	4.302	1.619

Table 16. Average disparities for the improved model.

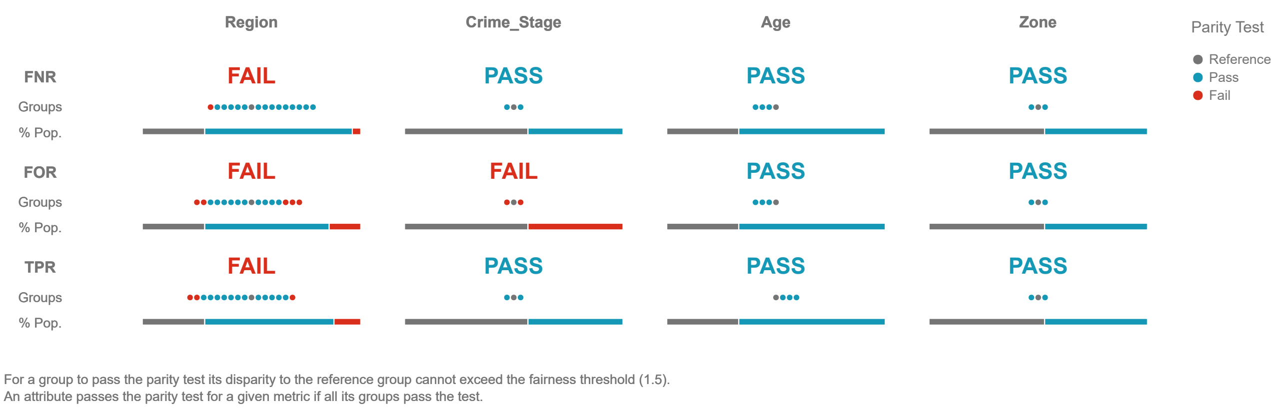

Figure 15. Model assessment for the reference group.

Figure 16. Model assessment for the major group.

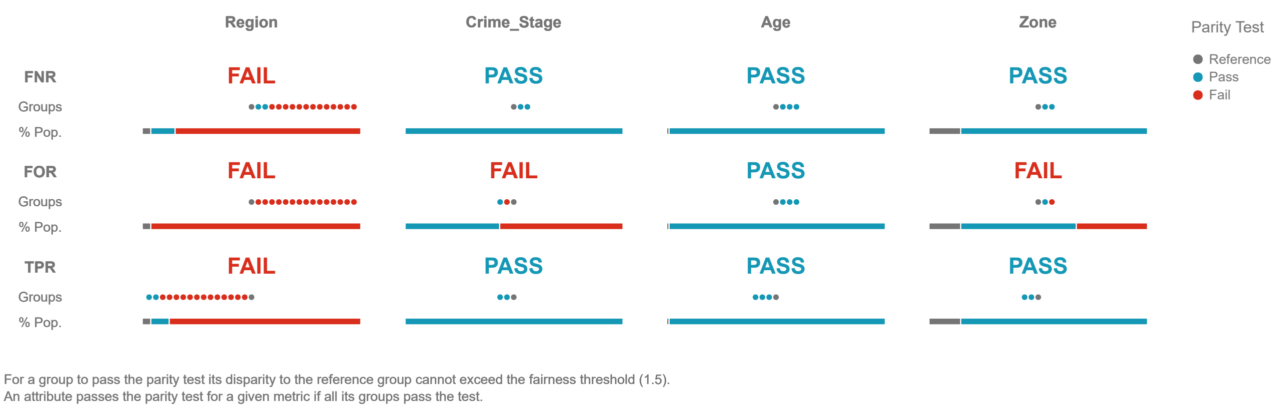

Figure 17. Model assessment for the min metrics group.

Attribute Value	TPR	FNR	FOR	FDR
Antofagasta	0.4938	0.5062	0.3154	0.2593
Arica y Parinacota	0.2581	0.7419	0.3407	0.5152
Atacama	0.5946	0.4054	0.0380	0.6857
Aysén	0.5556	0.4444	0.2000	0.6512
Bio Bío	0.4768	0.5232	0.1535	0.3543
Coquimbo	0.4136	0.5864	0.1750	0.6030
La Araucanía	0.3571	0.6429	0.2549	0.5167
Libertador Bernardo O'Higgins	0.3318	0.6682	0.1592	0.4779
Los Lagos	0.2292	0.7708	0.4964	0.3245
Los Ríos	0.3585	0.6415	0.1604	0.5250
Magallanes y Antártica Chilena	0.2059	0.7941	0.2411	0.7083
Maule	0.3039	0.6961	0.2243	0.7490
Metropolitana	0.3853	0.6147	0.2043	0.2657
Tarapacá	0.5714	0.4286	0.1091	0.6923
Valparaíso	0.2802	0.7198	0.2040	0.5526
Ñuble	0.3571	0.6429	0.1989	0.7143

Table 17. Average bias metrics between attributes for the improved model.

Table 16 and Figures 15 to 17 shows that disparities do improve, but the magnitude of the change is not as profound as in the previous experiment. Specifically, FNR disparities were decreased by an average of 25.40%, while FOR disparities decreased by 7.04%. However, TPR disparities actually increased by 11.71%. The model is still unable to attain overall fairness or statistical parity. Nevertheless, every variable, with the exception of "Region", aligns with FNR parity. "Zone" and "Age" are the sole achievers of TPR parity, with "Age" being the only one that achieved FOR parity. Despite improvements, "Region" does not have any parity alignment, even though its average bias metrics between attributes were also improved overall, as the FNR decreased from 85% to 61.41%, its TPR went up to 38.58%, and its FOR went down to 21.72%~~, therefore aligning with the statement made previously: the model is not only fairer, but it is also more accurate than the first one~~. These results indicate a narrower improvement in selected fairness metrics rather than a complete improvement across all criteria.

4 Conclusion & Future work

The improvements made using AI ~~Fariness~~ Fairness 360 and Fairlearn were ~~mainly positive , as both experiments improved their overall fairness~~ positive for selected fairness metrics, especially FNR and some Aequitas disparity measures. Biases and disparities by attribute were reduced in several cases, although these changes were not sufficient to achieve statistical parity and/or fairness.

The changes also improved ~~the parity levels between our~~ some parity levels for our proxy protected variable: Region~~, which means that the~~. This means that selected inequalities between the different regions of the country were reduced, ~~thus minimising potential biases~~but it does not mean that regional bias was eliminated or that Region captures all legally or socially relevant protected attributes.

~~Most importantly, these changes did not have an overall negative impact on model performance~~The results also show a fairness–performance trade-off. In the first experiment we obtained a ~~slight reduction in both~~ reduction in accuracy and precision ~~, but in~~ while improving recall and FNR. In the second experiment~~the model actually improvedin terms of performance.~~, recall and F1 improved, but accuracy and precision decreased. These results are useful for auditing and mitigation, but they are not enough to justify operational legal use without further validation.

But this prompts the question: to what extent do the models contribute to addressing these biases and inequalities? The response is likely not much in comparison to the data that is used for training. This is the primary conclusion drawn from these two experiments: the level of fairness depends more on the data itself than on the trained model. Essentially, as the model learns from the data patterns, it is ~~impossible~~ difficult to train something that disregards these issues. Historical judicial labels can encode legal, social, and institutional patterns that mitigation techniques cannot fully separate from the prediction target.

The work done by this and other frameworks is mainly to mitigate these problems, not to avoid them altogether, as they can be applied to a multitude of problems. If something like this is wanted or needed, it could be done in two ways: one is to adapt the way in which data that is already stored in a database is collected. This mainly involves creating more complex queries that already take these ethical aspects into account, although this is usually impractical. On the other hand, it can be beneficial to utilise data cleansing, quality checks, and data augmentation methods to minimise ethical concerns. One such approach would be to implement sampling techniques or generate synthetic data (e.g. SMOTE), although the obvious problem with this is that it can introduce noise into the model.

However, this is just one aspect of the issue. Quantity is crucial, as it is usually needed a dataset with a substantial number of entries for training, as well as having multiple variables that can potentially be utilised for these purposes, including those for doing some feature engineering. In this instance, we lacked features that may have been advantageous for improving the prediction accuracy of the models in both experiments. In particular, the analysis lacks critical variables, including the sex of the accused, real age, previous cases, and other relevant case information such as days, expert reports, and degree of participation.

Ultimately, while some improvement techniques can enhance models, data remains the primary factor in developing a responsible and impartial model. Thus, altering hyperparameters or adjusting prediction thresholds may prove insufficient when the available data is lacking, as has been demonstrated by these experiments.

In the end, this ~~framework~~ workflow is designed to be versatile and applicable across various domains and datasets, particularly for classification problems that require ~~ethical considerations~~fairness auditing and documentation. While it ~~may~~ does not provide a ~~perfect~~ complete solution for every scenario, it serves as a ~~valuable tool for addressing ethical concerns during the development of new models~~practical tool for identifying measurable error disparities, testing mitigation strategies, documenting limitations, and informing whether a model should be further developed, restricted, or rejected.

Finally, as we continue to develop and improve our methodology for identifying and addressing biases in ML models, a number of possibilities emerge for future investigation and enhancement of our current framework.

1. **Test additional protected variables**: Due to data limitations, the only protected variable we could test was "Region". In the future, other variables could be analysed in order to further validate this ~~framework~~workflow. Variables such as ~~the~~ nationality, ethnicity, ~~or the socioeconomic status– among others –~~sex, socioeconomic status, real age, prior cases, and other relevant case features could prove useful for these kinds of analyses.

2. **Optimisation Techniques in Model Training**: There is potential in using regularization methods that take into account protected variables[49]. By incorporating fairness considerations into the optimisation process, it is expected that models will be able to be trained to intrinsically decrease bias and more accurately represent all aspects of the data. This would have a twofold benefit, improving prediction precision while also guaranteeing that the model does not unintentionally continue societal biases.

3. **Data Disparity Sampling Techniques**: Data forms the fundamental basis upon which ML models are built. However, discrepancies in data may create a notable source of prejudice, particularly when certain groups are underrepresented. To overcome this hurdle, we could examine and utilise advanced sampling techniques customised to the distinct needs of each use case. By guaranteeing that the training data is a more representative sample of the wider population, we can create models that are both more precise and impartial. These techniques may vary from oversampling underrepresented groups to more advanced methods of creating synthetic data, guaranteeing that models have enough data to learn from even if the historical datasets are biased.

4. **External and prospective validation**: Future work should evaluate whether the workflow remains stable under new cases, different time periods, and independent datasets. This is particularly important in legal domains, where historical labels can change with policy, jurisprudence, and institutional practice.

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Data availability

The codes and datasets used can be found in the following repository:
Repository: https://github.com/nsalazarv/ethical-ml-framework
~~Pretty~~ Petty theft data: https://github.com/nsalazarv/ethical-ml-framework/blob/main/experiments/pettyTheft/petty_theft_anon.csv
Drug ~~traffiking~~ trafficking data: https://github.com/nsalazarv/ethical-ml-framework/blob/main/experiments/drugTrafficking/drug_traffiking_anon.csv

Additional information

A ~~Drug traffiking experiment~~Sample Model Card

This appendix provides a concise Model Card template for documenting the type of models audited in this paper. It is not a deployment approval document; it is intended to make the model's intended use, limitations, and fairness evidence explicit.

Field	Description
Model details	Binary classifier for predicting whether a case has a favourable or unfavourable outcome for the defendant. The experiments use MLP and LightGBM implementations depending on the dataset.
Intended use	Retrospective audit, quality-control analysis, and methodological evaluation. The model is not intended to replace legal judgement or to determine whether a lawyer should defend a person.
Out-of-scope use	Sentencing, bail, eligibility, defence allocation, automated case triage without human review, or any direct decision about a defendant's rights or legal strategy.
Data	Historical DPP records for drug trafficking and petty theft cases between 2017 and 2022. The available features are limited and omit important variables such as real age, sex, prior cases, and several case-specific legal facts.
Label definition	Favourable and unfavourable outcomes are simplified labels derived from case outcomes. They do not establish legal correctness, quality of defence, or absence of historical bias.
Protected/proxy attribute	Region is used as a geographic proxy for fairness auditing because other protected attributes were unavailable. It should not be interpreted as a complete protected-attribute analysis.
Fairness metrics	FNR, FOR, TPR, FPR, and Aequitas disparity ratios by group. Mitigation is considered successful only for the specific metrics reported, not for all possible fairness criteria.
Known limitations	Synthetic age is not observed age; model predictions inherit limitations and potential biases in historical data; statistical tests are computed from manuscript-level aggregate confusion matrices rather than exact paired prediction files.
Monitoring recommendations	Recompute performance and disparity metrics on new data, compare against the documented baseline, audit label drift, and review any operational use with legal and domain experts before deployment.

Table 18. Sample Model Card for documenting the audited classification models.

B Drug trafficking experiment

B.1 Model performance

Model/ Dataset	Accuracy	Precision	Recall	F1 Score
Base Train	80%	100%	62%	77%
Base Test	78%	99%	59%	74%
Improved Train	67%	67%	76%	71%
Improved Test	65%	64%	79%	71%

Table 19. Model performance.

B.2 SHAP Values

Feature	Model	Avg SHAP	Avg Abs SHAP
Effective Hearings	Base	0.000027	0.000467
	Improved	0.000053	0.000554
Age (Uniform)	Base	-0.000581	0.004270
	Improved	-0.000063	0.001798
Region	Base	0.002293	0.010827
	Improved	0.000169	0.005365
Barrister	Base	0.000666	0.008143
	Improved	-0.000198	0.005142
Crime Stage	Base	0.000082	0.000082
	Improved	0.000159	0.000159
Foreigner	Base	-0.025404	0.309130
	Improved	-0.025476	0.304273

Table 20. Average and average absolute SHAP values for the base and improved models.

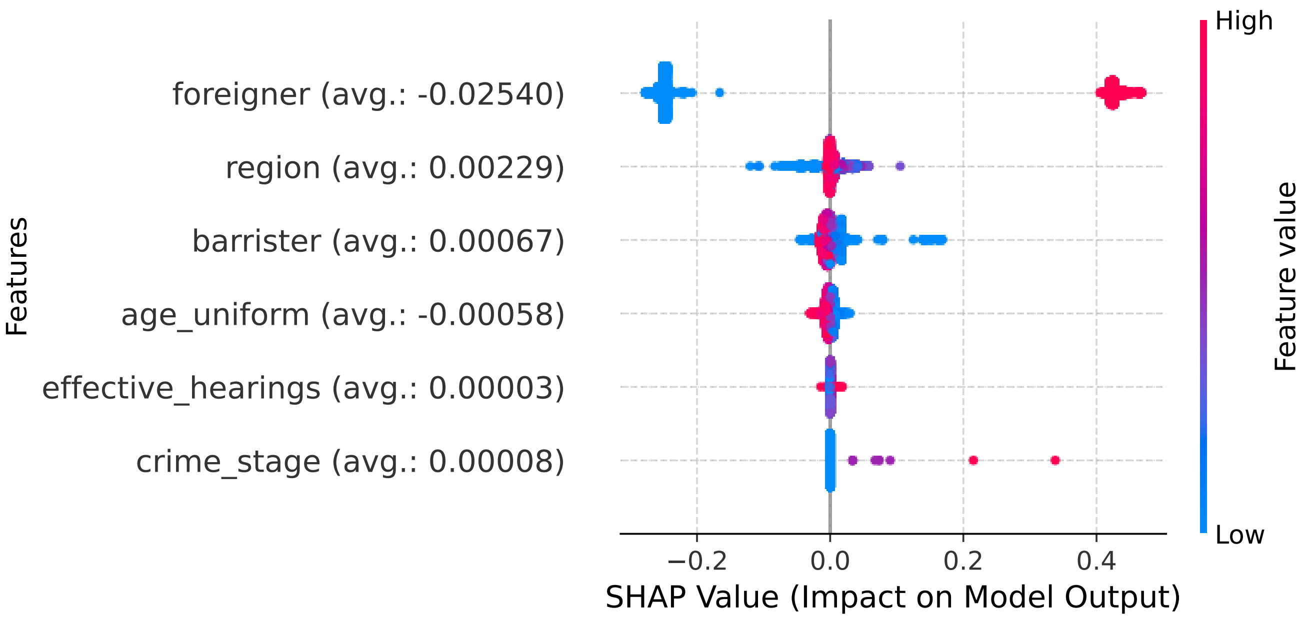

Figure 18. SHAP values distribution by feature (base model).

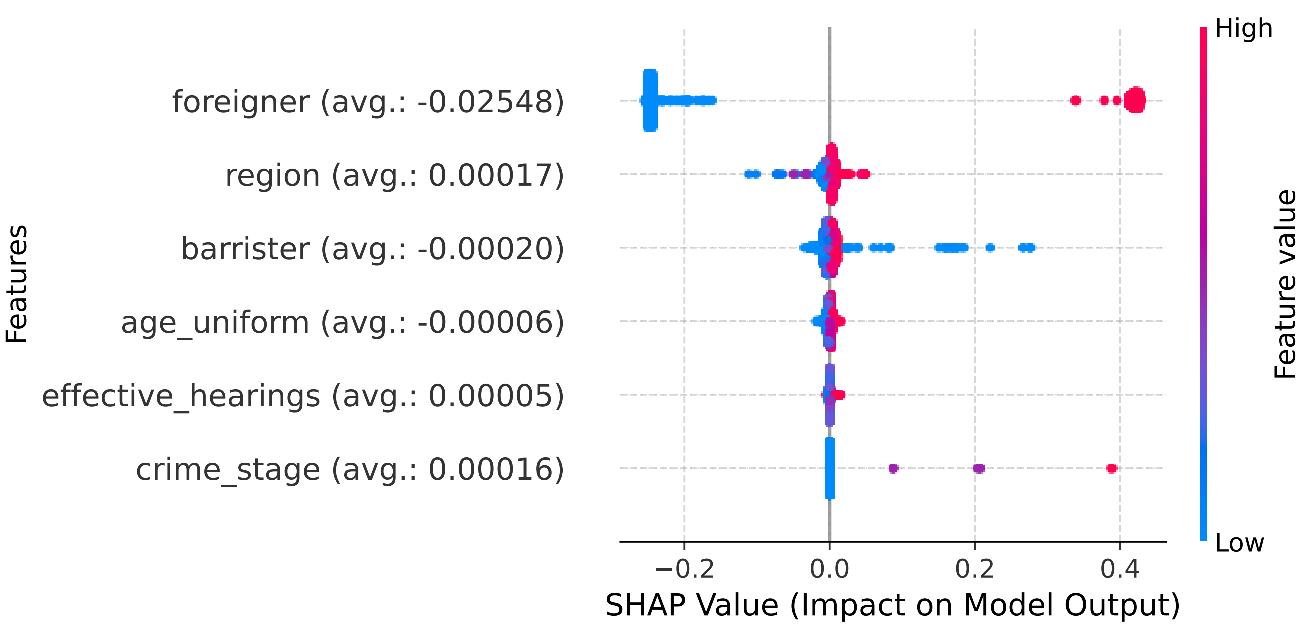

Figure 19. SHAP values distribution by feature (improved model).

B.3 Bias and Confusion Matrix

Model	FPR	FNR	FDR	FOR
Base	0.43%	40.51%	0.62%	31.92%
Improved	51.68%	20.70%	36.12%	33.06%

Table 21. Bias metrics between attributes.

Model	True Positives	True Negatives	False Positives	False Negatives
Base	1595	2316	10	1086
Improved	2126	1124	1202	555

Table 22. Confusion Matrix.

B.4 Disparities (Aequitas)

Base:

Assessment	FNR Disparities	FOR Disparities	TPR Disparities
Reference Group	0.80	1.07	14.51
Major Group	7.50	1.65	13.50
Min. Metrics Group	9.14	1.66	21.10

Table 23. Average disparities for the base model.

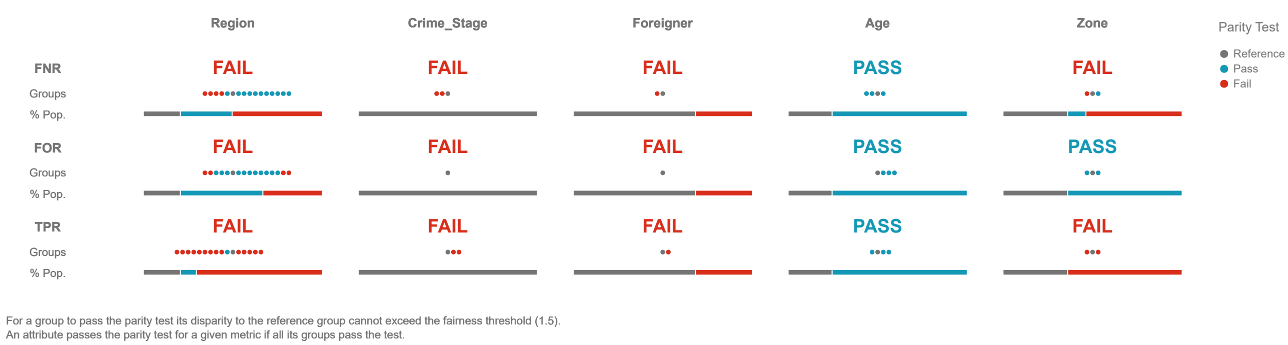

Figure 20. Model assessment for the reference group.

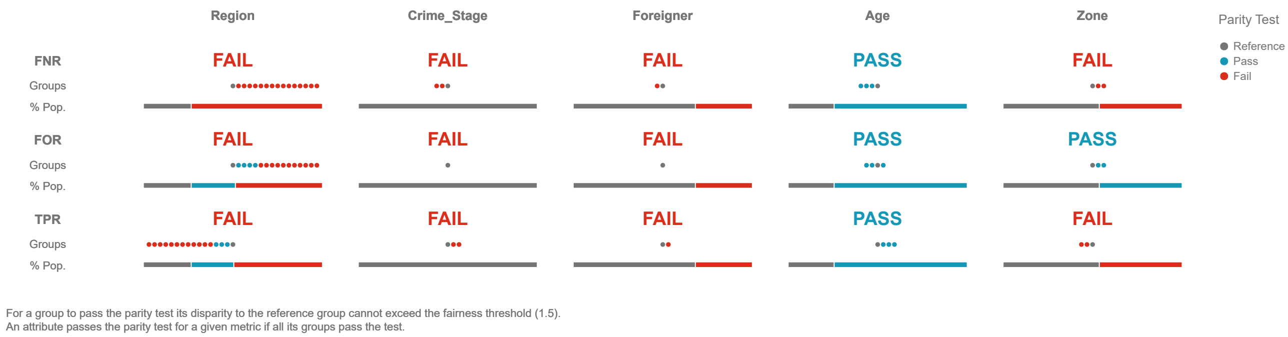

Figure 21. Model assessment for the major group.

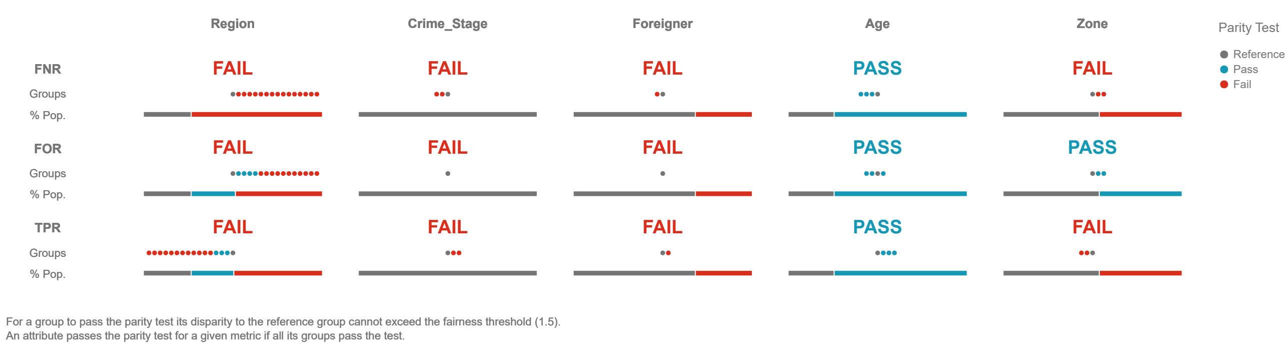

Figure 22. Model assessment for the min metrics group.

Attribute Value	TPR	FNR	FOR	FDR
Antofagasta	0.8213	0.1787	0.2167	0.0002
Arica y Parinacota	0.7276	0.2724	0.2571	0.0045
Atacama	0.6782	0.3218	0.2456	0.0002
Aysén	0.0002	1.0000	0.3889	-
Bio Bío	0.0139	0.9861	0.4057	0.0002
Coquimbo	0.2547	0.7453	0.5852	0.0002
La Araucanía	0.0556	0.9444	0.3238	0.0002
Libertador Bernardo O'Higgins	0.0566	0.9434	0.3788	0.4000
Los Lagos	0.0227	0.9773	0.3554	0.0002
Los Ríos	0.0002	1.0000	0.1818	-
Magallanes y Antártica Chilena	0.0909	0.9091	0.3448	0.0002
Maule	0.0156	0.9844	0.5000	0.0002
Metropolitana	0.1554	0.8446	0.3067	0.0678
Tarapacá	0.9377	0.0623	0.1720	0.0002
Valparaíso	0.1117	0.8883	0.3939	0.1250
Ñuble	0.0833	0.9167	0.4000	0.0002

Table 24. Average bias metrics between attributes for the base model.

Improved:

Assessment	FNR Disparities	FOR Disparities	TPR Disparities
Reference Group	1.15	1.41	0.99
Major Group	0.90	0.95	1.05
Min. Metrics Group	2.05	1.55	1.31

Table 25. Average disparities for the improved model.

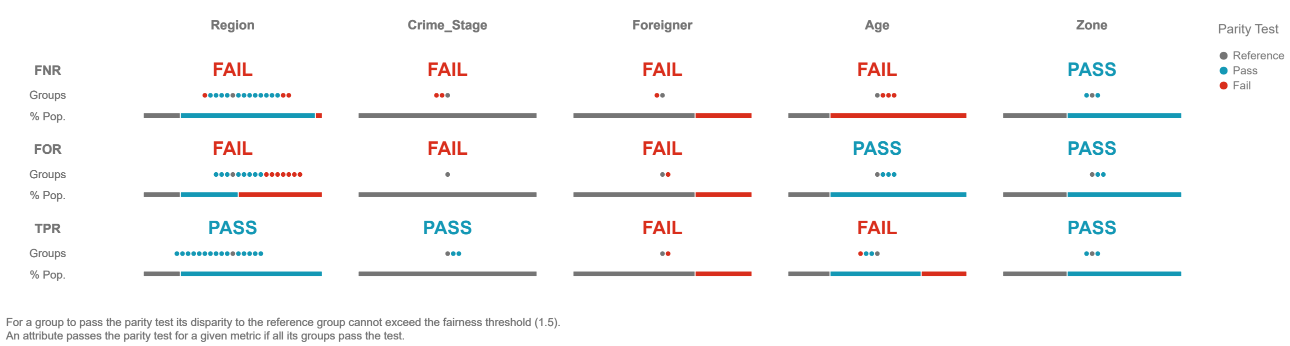

Figure 23. Model assessment for the reference group.

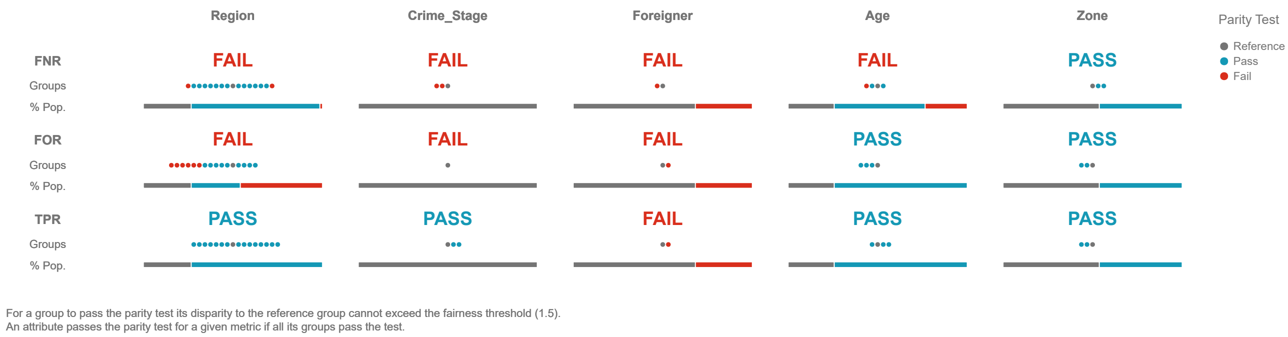

Figure 24. Model assessment for the major group.

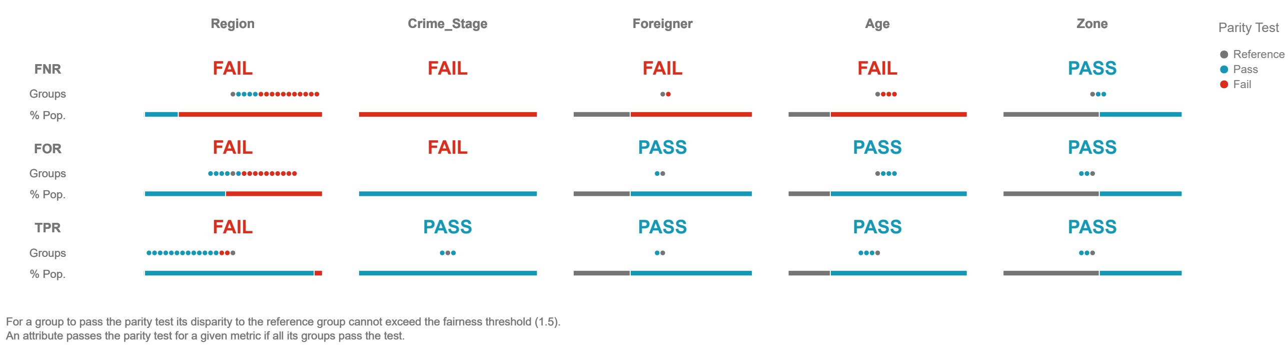

Figure 25. Model assessment for the min metrics group.

Attribute Value	TPR	FNR	FOR	FDR
Antofagasta	0.7113	0.2887	0.3088	0.0002
Arica y Parinacota	0.7309	0.2691	0.2588	0.0265
Atacama	0.7356	0.2644	0.2421	0.1795
Aysén	0.6429	0.3571	0.5556	0.6667
Bio Bío	0.7500	0.2500	0.3750	0.5781
Coquimbo	0.6887	0.3113	0.6111	0.3241
La Araucanía	0.6667	0.3333	0.5217	0.7143
Libertador Bernardo O'Higgins	0.5849	0.4151	0.4400	0.6437
Los Lagos	0.5455	0.4545	0.3636	0.6418
Los Ríos	0.5000	0.5000	0.2353	0.8519
Magallanes y Antártica Chilena	0.8182	0.1818	0.2857	0.6087
Maule	0.6719	0.3281	0.5526	0.5169
Metropolitana	0.7232	0.2768	0.2715	0.6196
Tarapacá	0.6877	0.3123	0.5103	0.0002
Valparaíso	0.6436	0.3564	0.4295	0.5856
Ñuble	0.7917	0.2083	0.3571	0.5581

Table 26. Average bias metrics between attributes for the improved model.

C Petty theft experiment

C.1 Model performance

Model/ Dataset	Accuracy	Precision	Recall	F1 Score
Base Train	78%	77%	33%	46%
Base Test	76%	67%	27%	38%
Improved Train	77%	63%	44%	52%
Improved Test	74%	53%	34%	42%

Table 27. Model performance.

C.2 SHAP Values

Feature	Model	Avg SHAP	Avg Abs SHAP
Effective Hearings	Base	-0.026311	0.379225
	Improved	-0.029706	0.376587
Age (Uniform)	Base	0.000838	0.005024
	Improved	0.001159	0.015643
Region	Base	0.023420	0.185918
	Improved	0.014002	0.183294
Barrister	Base	-0.018388	0.059781
	Improved	-0.012151	0.084927
Crime Stage	Base	0.011579	0.167150
	Improved	0.013751	0.185868

Table 28. Average and average absolute SHAP values for the base and improved models.

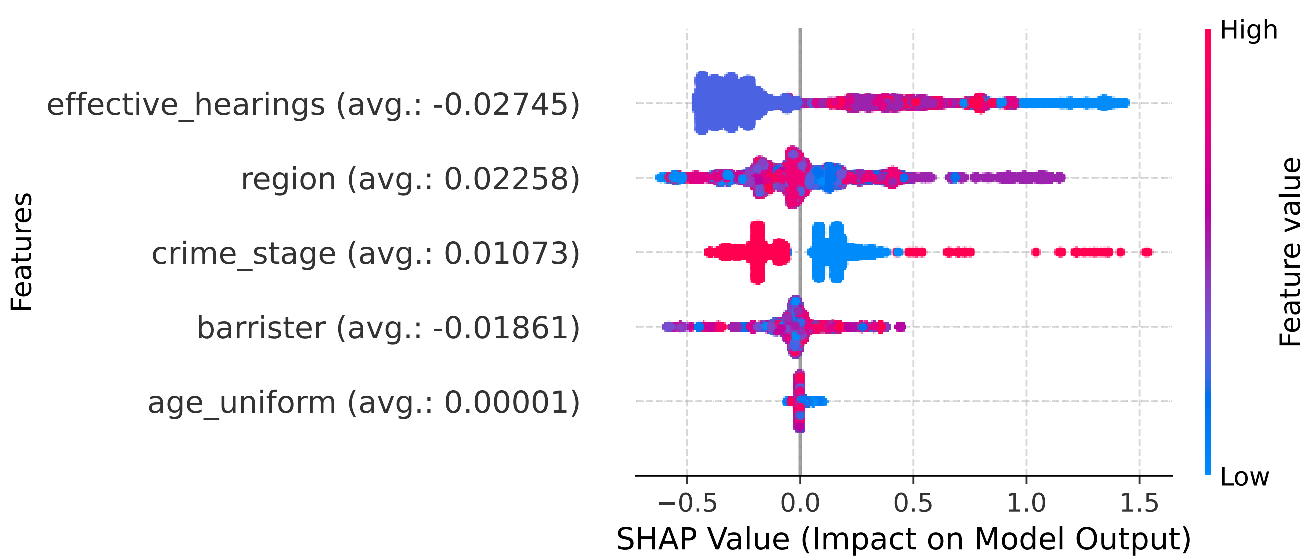

Figure 26. SHAP values distribution by feature (base model).

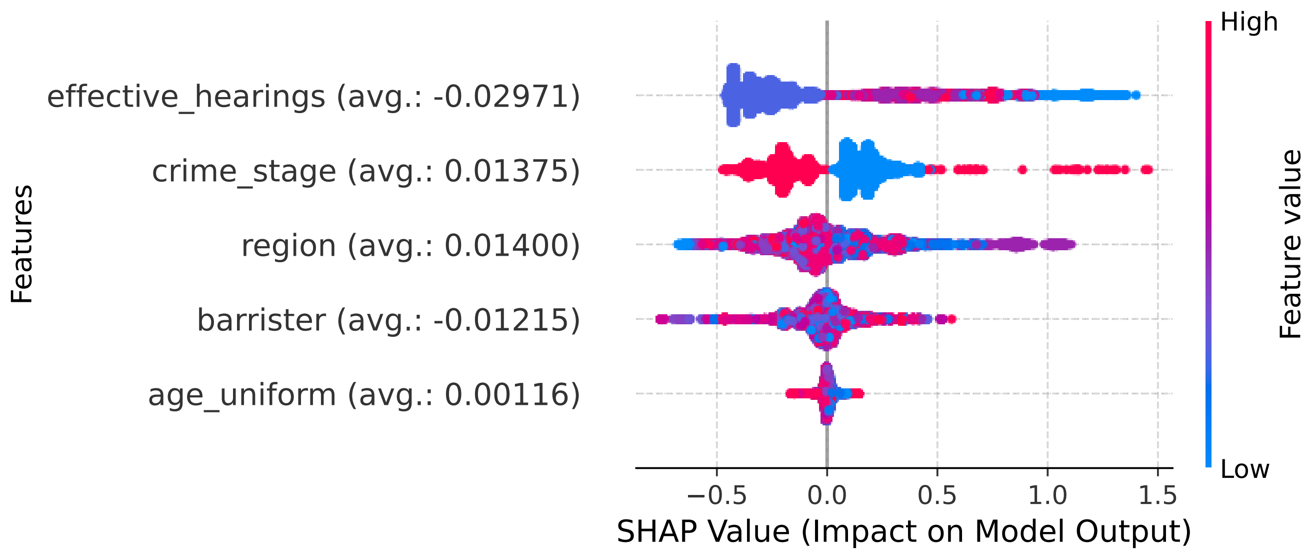

Figure 27. SHAP values distribution by feature (improved model).

C.3 Bias and Confusion Matrix

Model	FPR	FNR	FDR	FOR
Base	4.92%	73.29%	32.90%	22.47%
Improved	11.14%	66.03%	46.57%	21.84%

Table 29. Bias metrics between attributes.

Model	True Positives	True Negatives	False Positives	False Negatives
Base	875	8282	429	2401
Improved	1113	7741	970	2163

Table 30. Confusion Matrix.

C.4 Disparities (Aequitas)

Base:

Assessment	FNR Disparities	FOR Disparities	TPR Disparities
Reference Group	1.27	1.15	0.66
Major Group	1.27	1.15	0.66
Min. Metrics Group	2.17	2.89	6.26

Table 31. Average disparities for the base model.

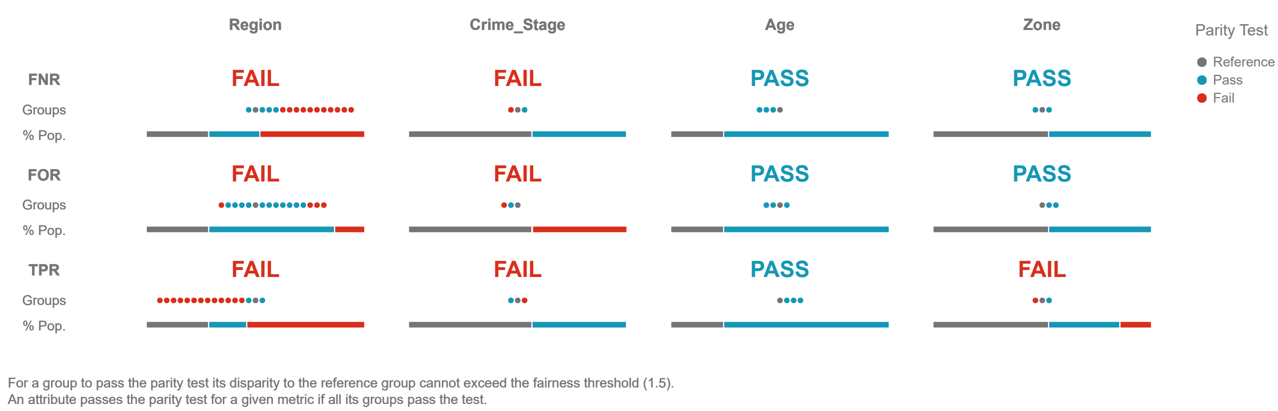

Figure 28. Model assessment for the reference group.

Figure 29. Model assessment for the major group.

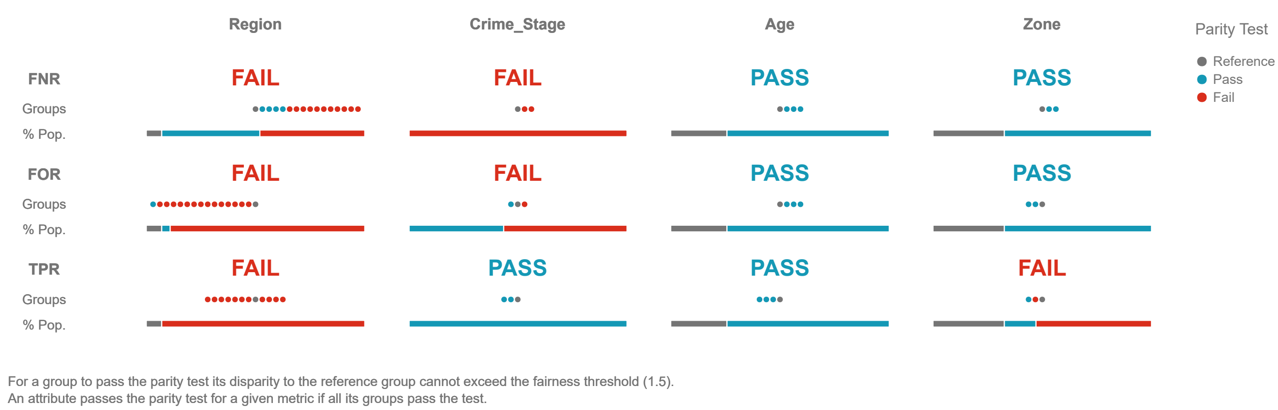

Figure 30. Model assessment for the min metrics group.

Attribute Value	TPR	FNR	FOR	FDR
Antofagasta	0.1235	0.8765	0.4201	0.3333
Arica y Parinacota	0.0806	0.9194	0.3519	0.1667
Atacama	0.2432	0.7568	0.0625	0.4706
Aysén	0.0002	1.0000	0.2700	1.0000
Bio Bío	0.3642	0.6358	0.1752	0.2949
Coquimbo	0.0890	0.9110	0.2180	0.5854
La Araucanía	0.1232	0.8768	0.2821	0.1935
Libertador Bernardo O'Higgins	0.0841	0.9159	0.1943	0.2800
Los Lagos	0.4921	0.5079	0.4913	0.4267
Los Ríos	0.2642	0.7358	0.1674	0.2632
Magallanes y Antártica Chilena	0.0294	0.9706	0.2538	0.8333
Maule	0.0637	0.9363	0.2208	0.1333
Metropolitana	0.4225	0.5775	0.1952	0.2644
Tarapacá	0.0002	1.0000	0.1728	-
Valparaíso	0.0002	1.0000	0.2411	1.0000
Ñuble	0.0002	1.0000	0.2231	-

Table 32. Average bias metrics between attributes for the base model.

Improved:

Assessment	FNR Disparities	FOR Disparities	TPR Disparities
Reference Group	1.00	1.08	1.00
Major Group	1.00	1.08	0.99
Min. Metrics Group	1.33	3.91	1.72

Table 33. Average disparities for the improved model.

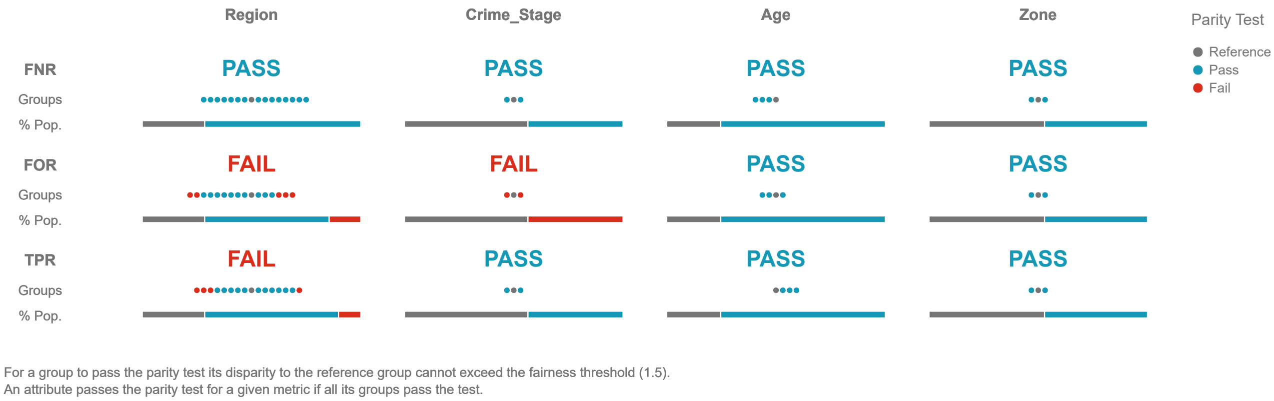

Figure 31. Model assessment for the reference group.

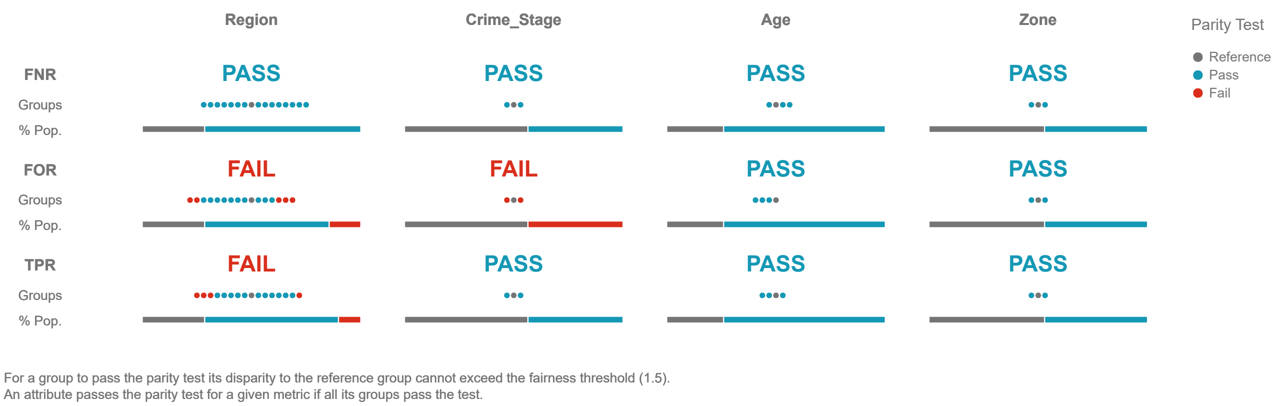

Figure 32. Model assessment for the major group.

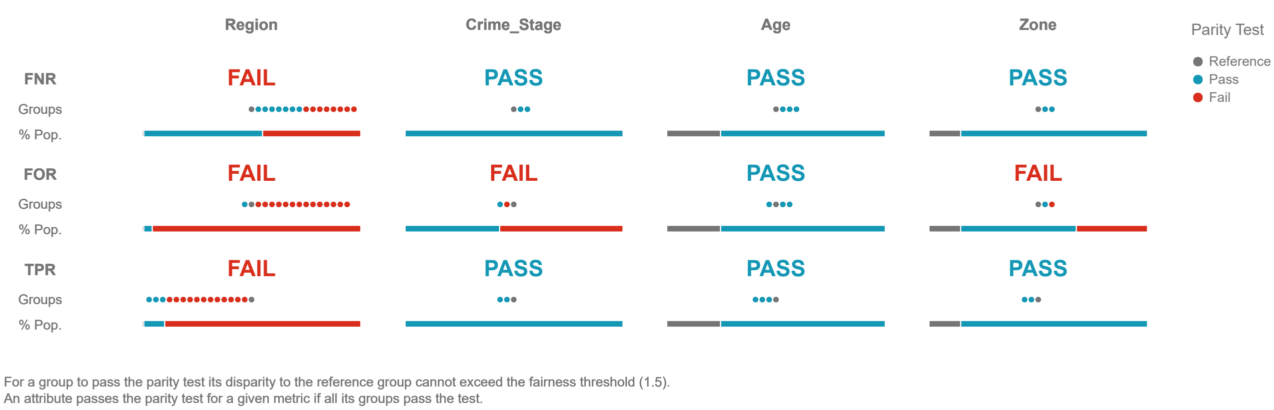

Figure 33. Model assessment for the min metrics group.

Attribute Value	TPR	FNR	FOR	FDR
Antofagasta	0.4198	0.5802	0.3381	0.2444
Arica y Parinacota	0.2258	0.7742	0.3529	0.5625
Atacama	0.5405	0.4595	0.0412	0.6154
Aysén	0.5185	0.4815	0.2031	0.6410
Bio Bío	0.4172	0.5828	0.1664	0.3505
Coquimbo	0.4084	0.5916	0.1725	0.5761
La Araucanía	0.3424	0.6576	0.2570	0.5123
Libertador Bernardo O'Higgins	0.3364	0.6636	0.1574	0.4545
Los Lagos	0.2045	0.7955	0.5050	0.3546
Los Ríos	0.3774	0.6226	0.1542	0.4737
Magallanes y Antártica Chilena	0.1765	0.8235	0.2435	0.7143
Maule	0.2696	0.7304	0.2221	0.7368
Metropolitana	0.3688	0.6312	0.2084	0.2684
Tarapacá	0.5714	0.4286	0.1111	0.7037
Valparaíso	0.3297	0.6703	0.1961	0.5522
Ñuble	0.3393	0.6607	0.1958	0.6935

Table 34. Average bias metrics between attributes for the improved model.